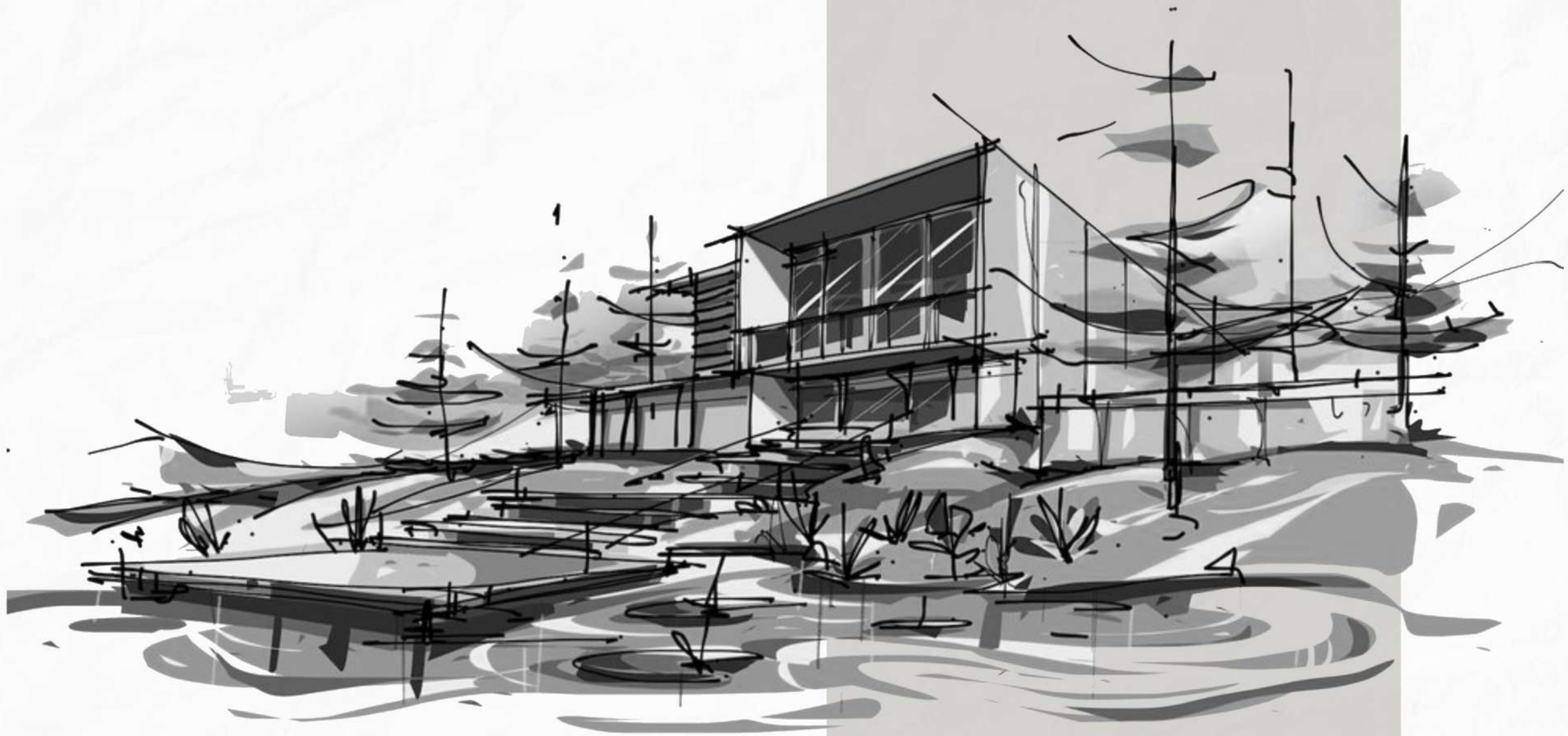
ARCHITECTURAL PORTFOLIO



Soham Khetan

Hello,

I'm Soham - an architecture student passionate about designing minimal and meaningful spaces. I'm drawn to clean, functional forms and restrained color palettes that create calm, intentional environments. I believe that good design is not only aesthetic but also responsible, which is why I'm deeply interested in sustainable architecture. I strive to create spaces that are efficient, environmentally conscious, and adaptable to future needs.

Beyond academics, I'm an avid gamer, which fuels my interest in immersive environments and spatial storytelling. I'm also enthusiastic about sports, which has taught me discipline, teamwork, and resilience - qualities I bring into my design process.

Travel inspires me with diverse cultures, architecture, and ways of living, constantly expanding my perspective as a designer.

sohamkhetan@gmail.com

+91 9356846922

@soham_ktan

18/10/2003

Chandrapur, Maharashtra

https://issuu.com/sohamkhetan

Soham Khetan



Education

High School Eduction (2019) (10th)
Narayana Vidyalayam, Chandrapur
Passed with 89.2%

Intermediate Education (2021) (10+2)
Narayana Vidyalayam, Chandrapur
Passed with 87.2%

Under Graduation (From 2021)
Sri Sri University, Cuttack
Currently a 5th year student

Academic Projects

- **1st Year** - 2nd Sem - Students Den
- **2nd Year** - 3rd Sem - Sculptor's Residence
  4th Sem - School For All
- **3rd Year** - 5th Sem - Science Park
  6th Sem - Vertical Housing
- **4th Year** - 7th Sem - Campus Planning
  8th Sem - Market Redevelopment

Skills

Manual Skills


Sketching


Drafting


Model Making

Software Skills


AutoCad


SketchUp


Photoshop


Enscape


Lumion


E-TABS


MS-Office

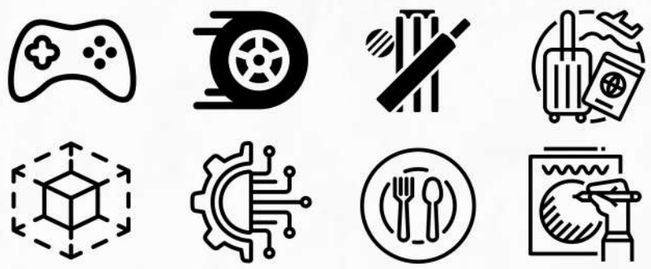
Languages

English
हिन्दी
मराठी
मारवाड़ी
ଓଡ଼ିଆ

Workshops

Photography Workshop - Ar. Shakti Nanda (2022)
Glass Painting Workshop - Ar. Soumyashree Mohanty (2023)
BIM Awareness - BMEDGE and CYPE (2023)
Design For Greater Effeciency Workshop (2024)
Dance-Music-Architecture Workshop - Ar. Anita Kulkarni (2024)
Sense n Sensibility 15.0 - ACEDGE (2024)

Interests



Competitions

Vertical Design Studio - 1st Runner Up (2022)
Transparence 17.0 - Participated (2022)
ANDC - Top 60 (2023)
IIA Mask Making Competition - 1st Prize (2023)
Zonal Nasa Convention - 2nd Runner up(D24)(2023)
Evolve Design Compeition - 1st Runner Up(2024)

CONTENTS

01 Crest Harbour
VERTICAL HOUSING (PAGE 6-9)

02 Monjee University of Managment
CAMPUS PLANNING (PAGE 10-13)

03 Unit One Market
MARKET REDEVELOPMENT (PAGE 14-17)

04 Evolve.io
INTERIOR DESIGN (PAGE 18-21)

05 Nirmaan
WORKING DRAWING (PAGE 22-23)

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ANDC 2023 (PAGE 24-25)

07 Competetion Work
VERTICAL DESIGN STUDIO 2023 (PAGE 26-27)

08 Miscellaneous
BASIC DESIGN (PAGE 28-29)

Crest Harbour

LOCATION - Hyderabad

SITE AREA - 24650 Sqm / 6.09Acres

SEMESTER - 6TH SEM

ABOUT - As urban populations surge and land becomes an increasingly limited resource, vertical housing emerges as a vital architectural solution to reconcile the tension between density and livability. This project, set in the evolving urban fabric of Hyderabad, envisions a vertical neighbourhood that transcends the limitations of traditional high-rise living by fostering community, safety, and inclusivity. Integrating structural innovation with climate-responsive strategies, the proposal seeks to redefine high-density housing as a harmonious blend of private comfort and collective belonging. This design journey focuses not only on vertical growth but also on nurturing the horizontal values of community, sustainability, and human-centric urbanism.



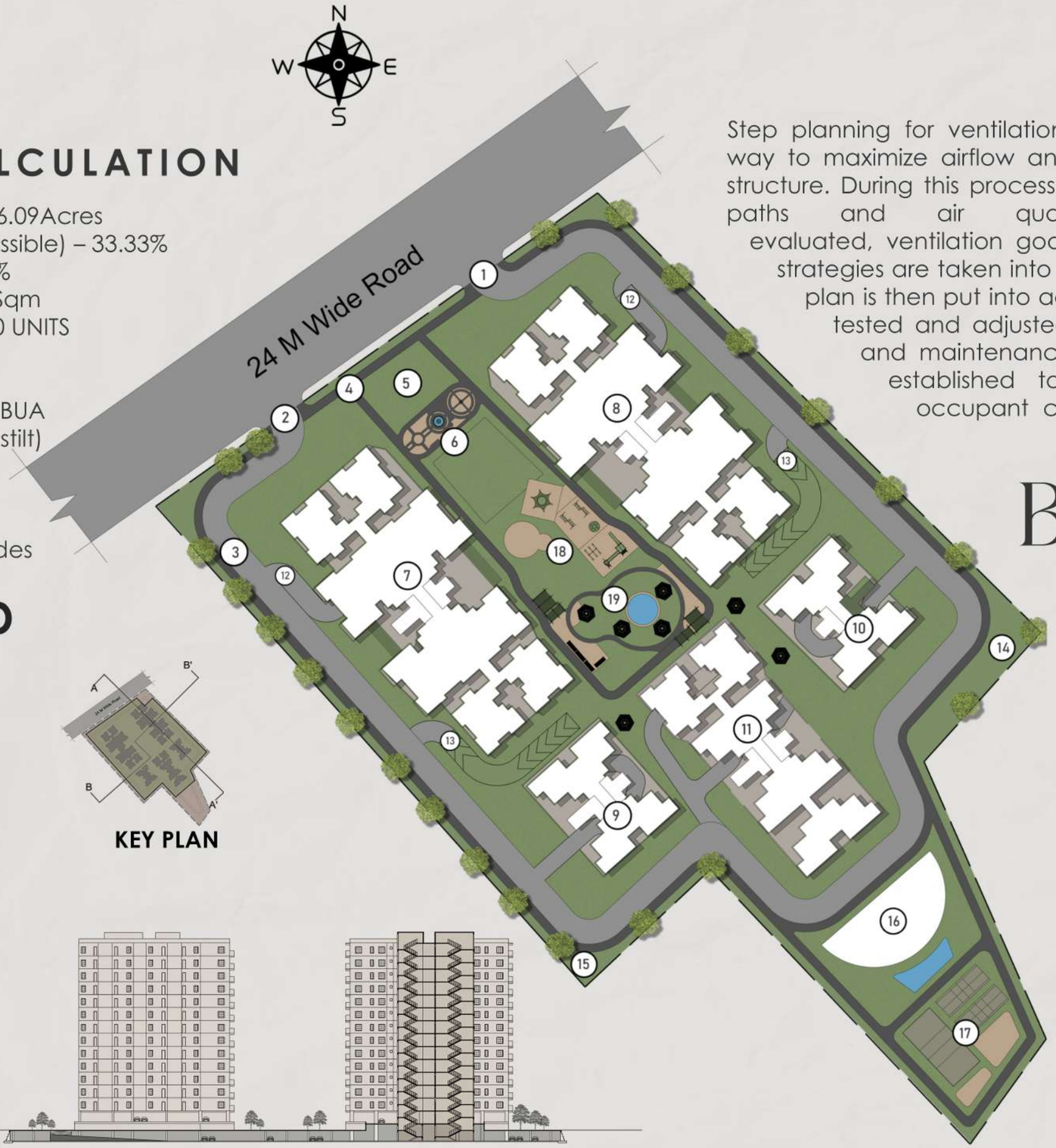
Site Plan

AREA CALCULATION

Site Area - 24650 Sqm or 6.09Acres
Ground coverage (Permissible) – 33.33%
Ground Coverage - 23.7%
Built Up Area - 81931.36 Sqm
Total Units Achieved - 400 UNITS
Permissible Height- 45 m
Achieved Height - 44.3m
Parking (HMDA) – 20% of BUA
Visitor's parking – 10% (In stilt)
Common Area - 25%
Totlot-10%
Amenities – 4%
Set backs – 13 m on all sides

LEGEND

1. SITE ENTRY/EXIT (1)
2. SITE ENTRY/EXIT (2)
3. SERVICE ROAD
4. PEDESTRIAN PATHWAY
5. LANDSCAPE GARDEN
6. FOUNTAIN
7. BLOCK 1
8. BLOCK 2
9. BLOCK 3
10. BLOCK 4
11. BLOCK 5
12. STILT RAMP
13. BASEMENT RAMP
14. PLUMBING SERVICE
15. ELECTRICAL SERVICE
16. CLUBHOUSE
17. PLAYING COURTS
18. TOTLOT AREA
19. OPEN GAZEBO

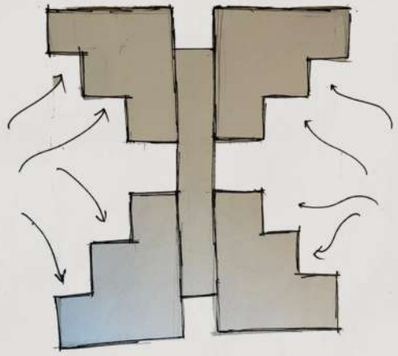


KEY PLAN

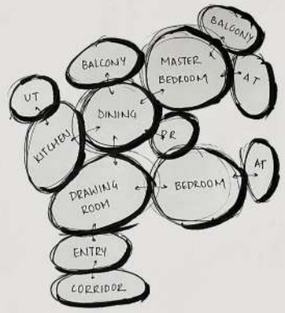
SITE SECTION BB'

Concept

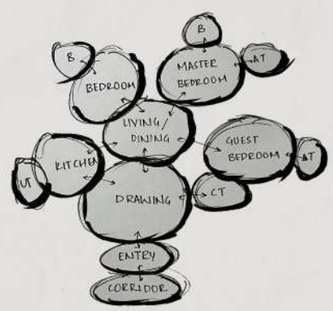
Step planning for ventilation is a methodical way to maximize airflow and quality inside a structure. During this process, the existing wind paths and air quality are first evaluated, ventilation goals are set, design strategies are taken into consideration. The plan is then put into action, the system is tested and adjusted, and monitoring and maintenance procedures are established to ensure optimal occupant comfort over time.



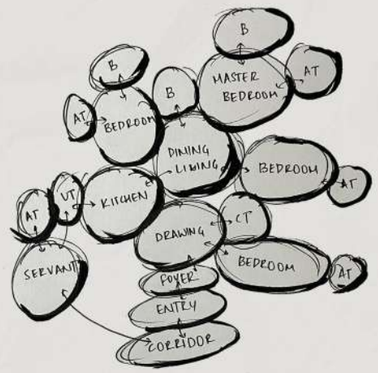
Bubble Diagram



2 BHK

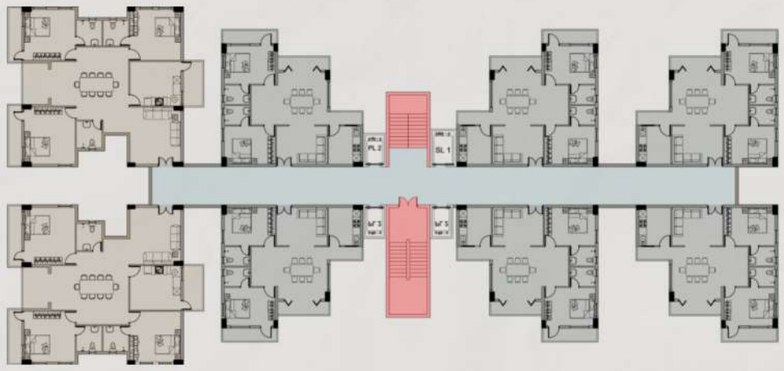


3 BHK



4 BHK

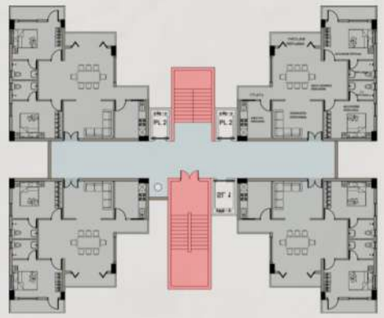
Floor Plates



BLOCK 5



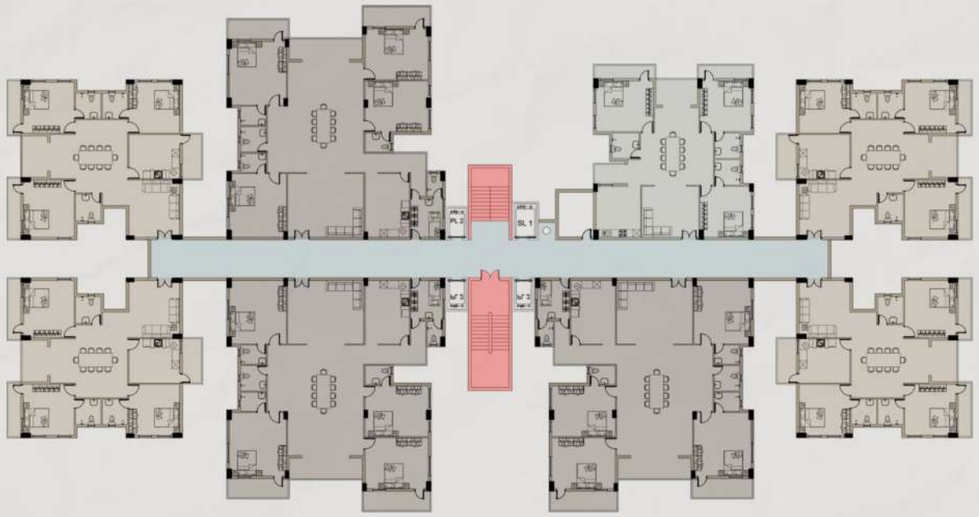
GROUND COVERAGE- 1104.48 SQM
BUILT UP AREA- 15462.72 SQM
NO. OF FLOORS- S+13
NO. OF UNITS IN 1 FLOOR- 8
TOTAL NO. OF UNITS- 104



BLOCK 3&4



GROUND COVERAGE- 506.38 SQM
BUILT UP AREA- 7089.32 SQM
NO. OF FLOORS- S+13
NO. OF UNITS IN 1 FLOOR- 4
TOTAL NO. OF UNITS- 52



LEGEND

- 3BHK(A)
- 3BHK(B)
- 4BHK
- CORRIDOR
- 2BHK

BLOCK 1&2



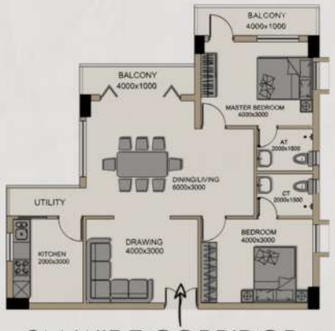
GROUND COVERAGE- 1867.5 SQM
BUILT UP AREA- 26145 SQM
NO. OF FLOORS- S+13
NO. OF UNITS IN 1 FLOOR-8
TOTAL NO. OF UNITS- 104







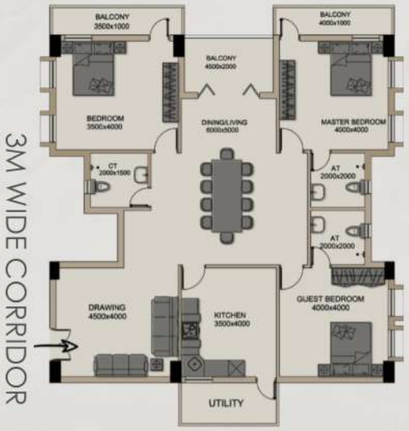
Unit Plans



2 B H K



4 B H K SUPREME



3 B H K (A)



3 B H K (B)

Sections



SECTION AA'



SECTION BB'





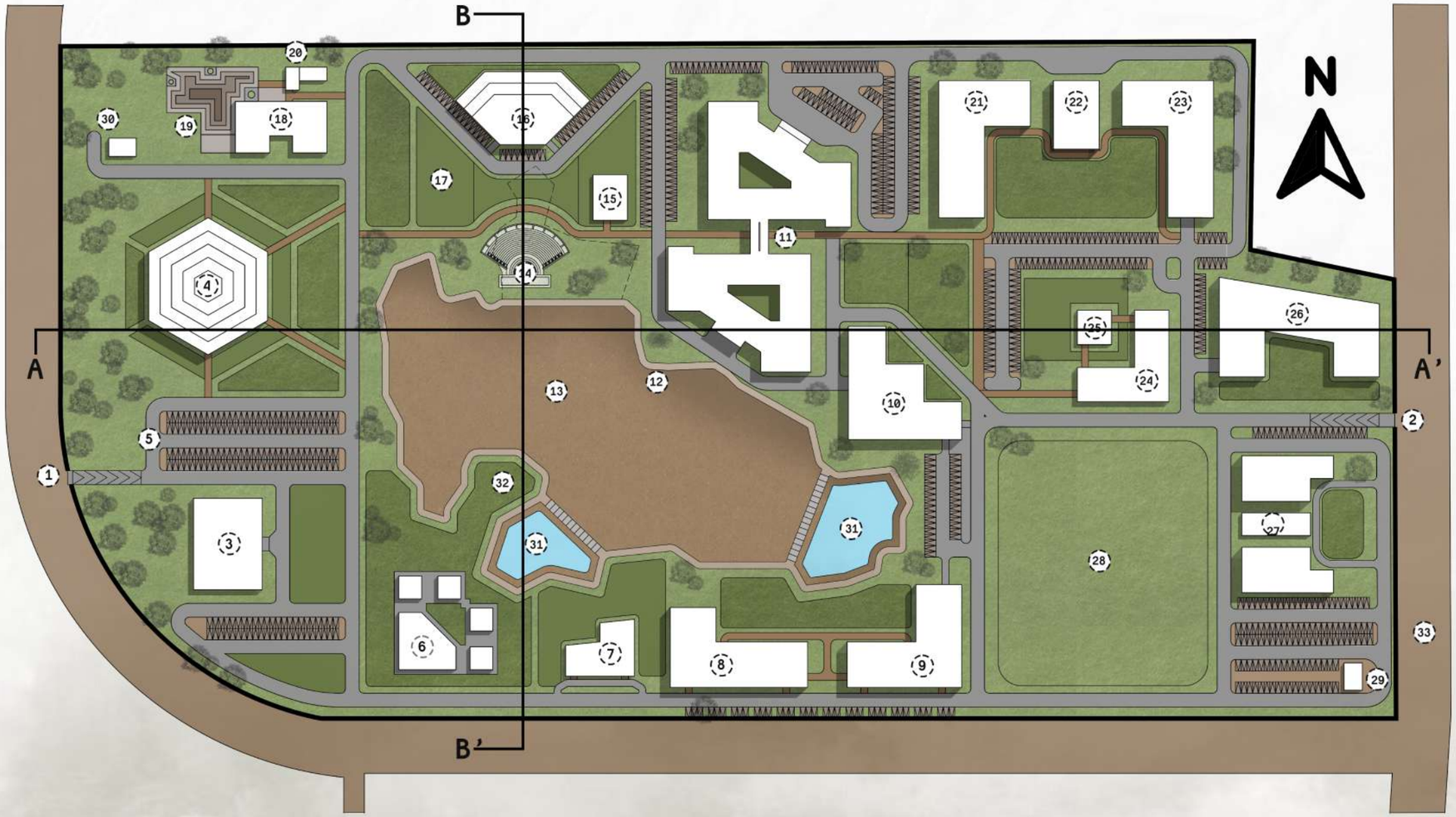


Monjee University of Management

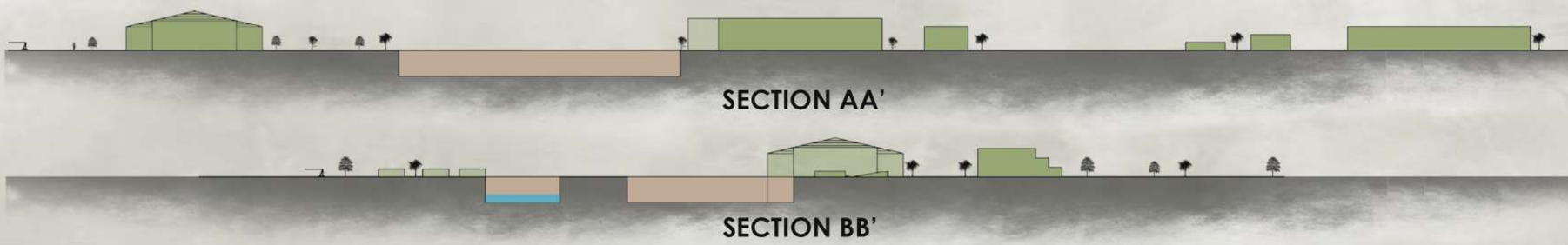
LOCATION - BHUBANESWAR
SITE AREA - 1,61,778 SQ.M/ 39.97 ACRES
SEMESTER - 7th SEM
ABOUT - The proposed design project envisions a comprehensive and future-ready campus for Narsee Monjee University of Management in Bhubaneswar, Odisha. Spread over 40 acres in the institutional corridor of Info Valley-II. The campus aims to integrate academic excellence with sustainability, inclusivity, and user-centric planning. Catering to diverse disciplines including Management, Engineering, Architecture, and Humanities, the project challenges the architect to create a vibrant learning environment that supports innovation, collaboration, and well-being. Rooted in contextual relevance and guided by national regulatory standards, the campus will serve as a model for contemporary educational design.



Site Plan



SITE SECTIONS



AREA STATEMENT

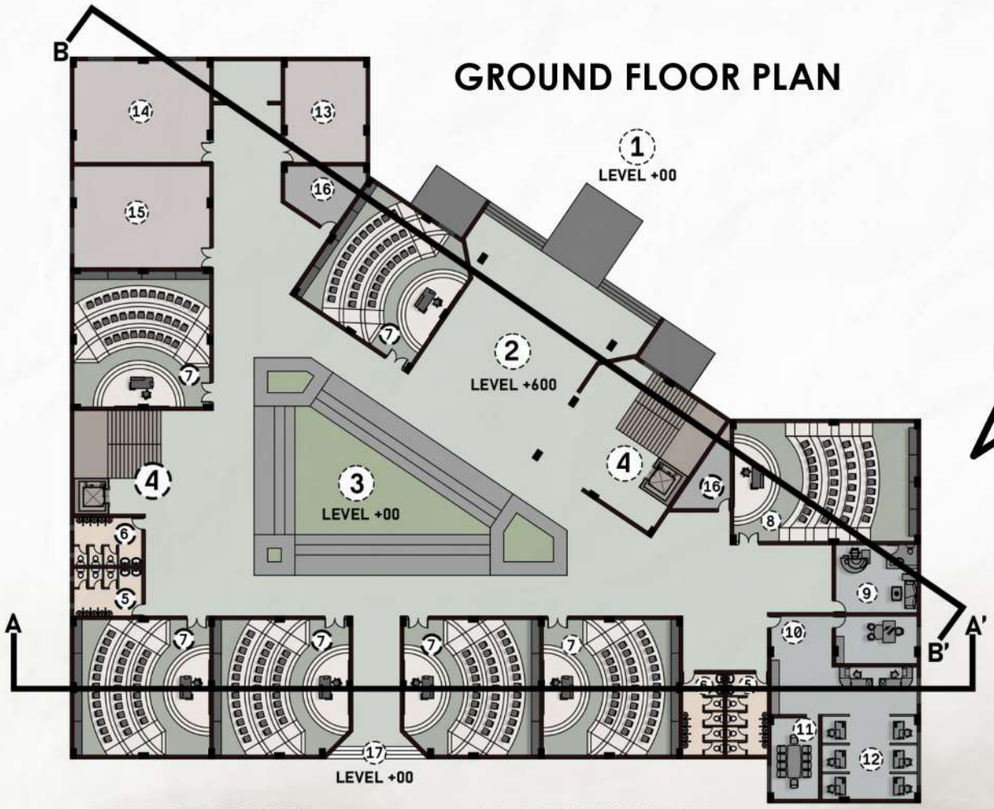
1. SITE AREA - 161778 SQ.M/ 39.97 ACRES
2. GROUND COVERAGE - 24498 SQ.M
3. BUILT UP AREA - 43824 SQ.M
4. PERMISSIBLE GROUND COVERAGE - 33%
5. ACHIEVED G.C. - 27.08%

LEGENDS

1. MAIN ENTRY/EXIT
2. SECONDARY ENTRY/EXIT
3. ADMIN BLOCK
4. AUDITORIUM
5. AUDITORIUM PARKING
6. GUEST HOUSES
7. VC RESIDENCE
8. ENGINEERING BLOCK
9. HUMANITIES BLOCK
10. ARCHITECTURE BLOCK
11. MANAGEMENT BLOCK
12. JOGGING TRACK
13. QUARRY
14. OPEN AIR THEATRE
15. HOSPITAL
16. CENTRAL LIBRARY
17. SENSORY PARK
18. INDOOR SPORTS COMPLEX
19. STEPPED OPEN SEATING
20. CAFE
21. BOYS HOSTEL
22. MESS
23. GIRLS HOSTEL
24. SHOPPING COMPLEX
25. CAFE
26. SCHOOL
27. FACULTY HOUSING
28. PLAY GROUND
29. STP
30. ELECTRICAL SERVICES
31. WATER PONDS
32. VIEW POINT
33. 24M WIDE ROAD

Floor Plans UNDER - GRADUATE

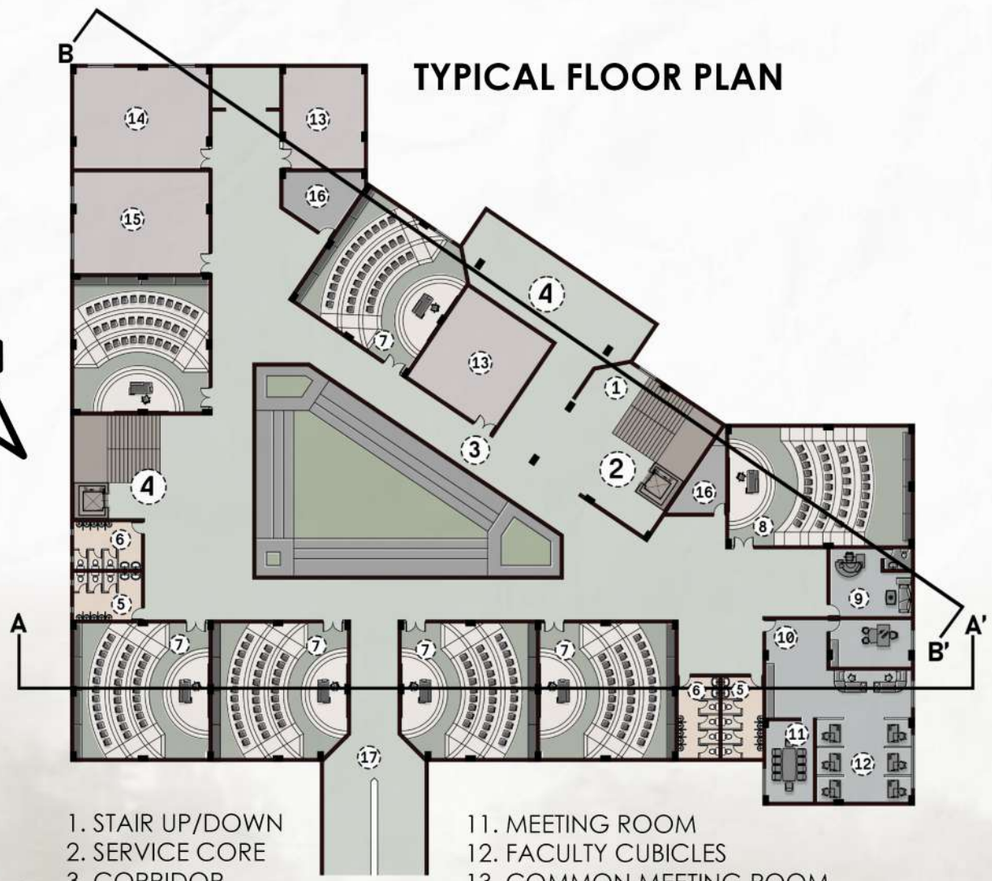
GROUND FLOOR PLAN



1. MAIN ENTRY/EXIT
2. ENTRANCE LOBBY
3. CENTRAL COURTYARD
4. SERVICE CORE
5. MALE TOILETS
6. FEMALE TOILETS
7. LECTURE HALL (TYPE 1)
8. LECTURE HALL (TYPE 2)
9. DEAN'S OFFICE
10. FACULTY AREA ENTRANCE
11. MEETING ROOM
12. FACULTY CUBICLES
13. COMMON MEETING ROOM
14. LANGUAGE ROOM
15. TUTORIAL ROOM
16. STORE ROOM
17. SECONDARY ENTRY/EXIT

N

TYPICAL FLOOR PLAN

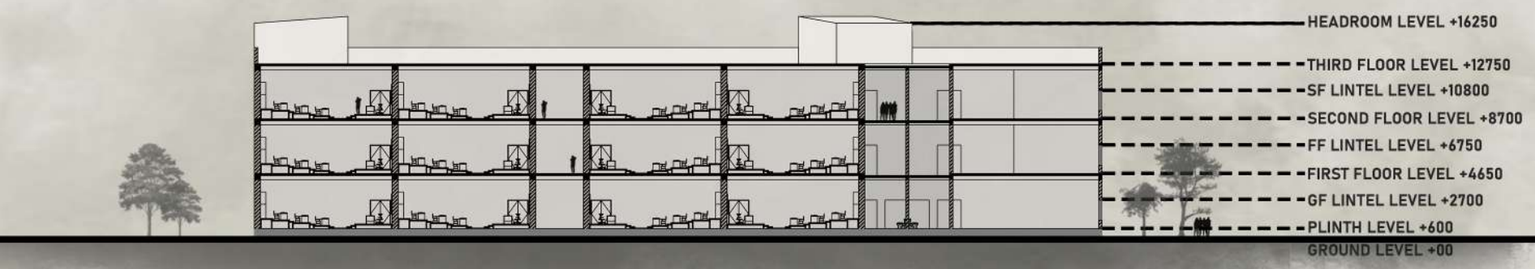


1. STAIR UP/DOWN
2. SERVICE CORE
3. CORRIDOR
4. OPEN SEATING
5. MALE TOILETS
6. FEMALE TOILETS
7. LECTURE HALL (TYPE 1)
8. LECTURE HALL (TYPE 2)
9. DEAN'S OFFICE
10. FACULTY AREA ENTRANCE
11. MEETING ROOM
12. FACULTY CUBICLES
13. COMMON MEETING ROOM
14. LANGUAGE ROOM
15. TUTORIAL ROOM
16. STORE ROOM
17. CONNECTING BRIDGE

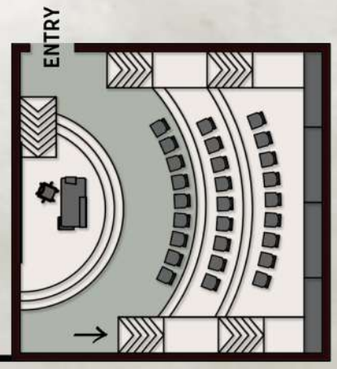
KEY PLAN



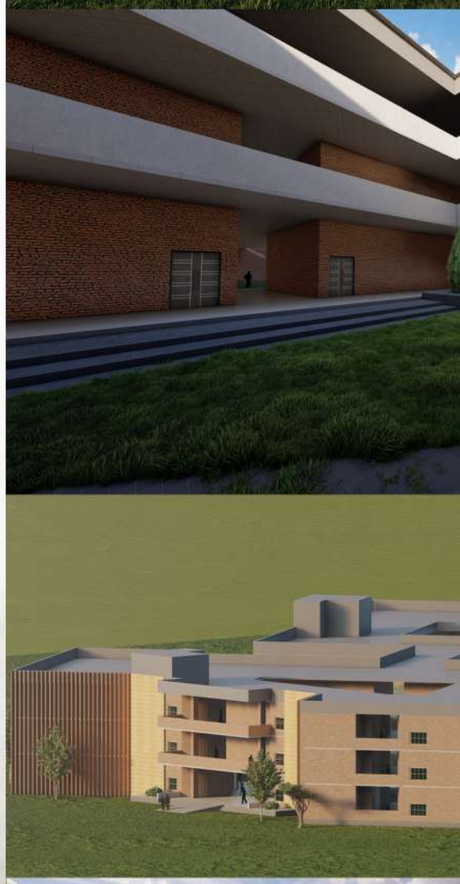
SECTION AA'



REFINED CLASSROOM DETAIL (BARRIER FREE DESIGN)



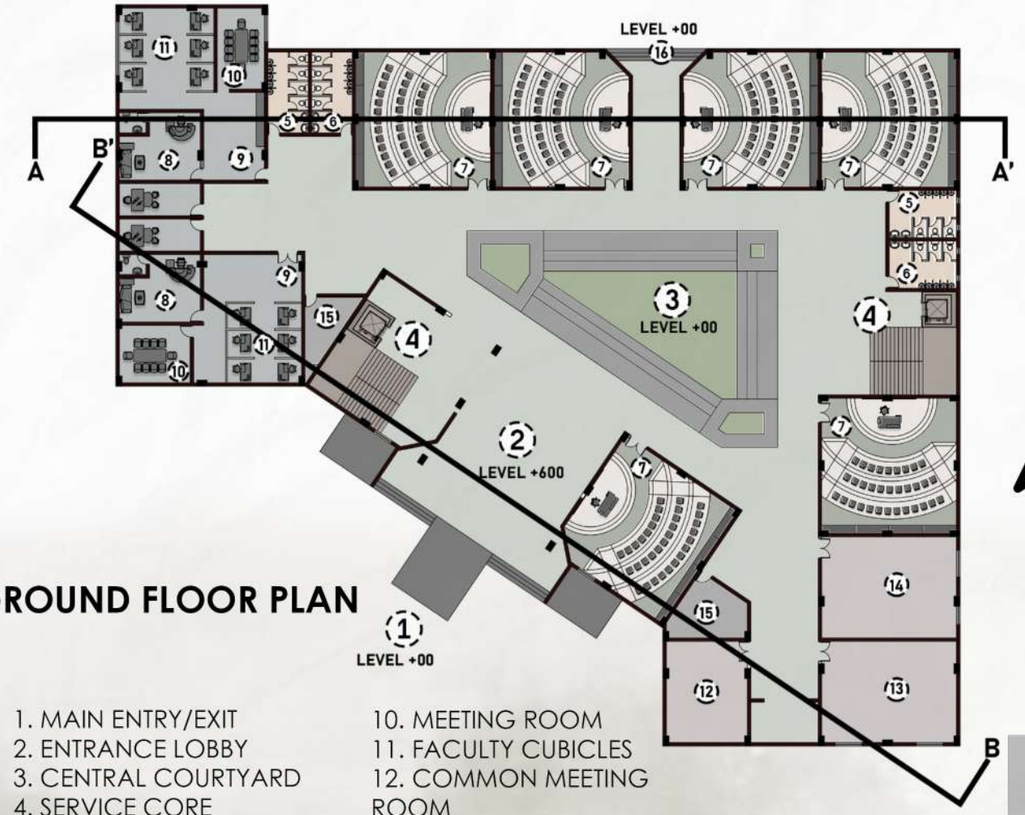






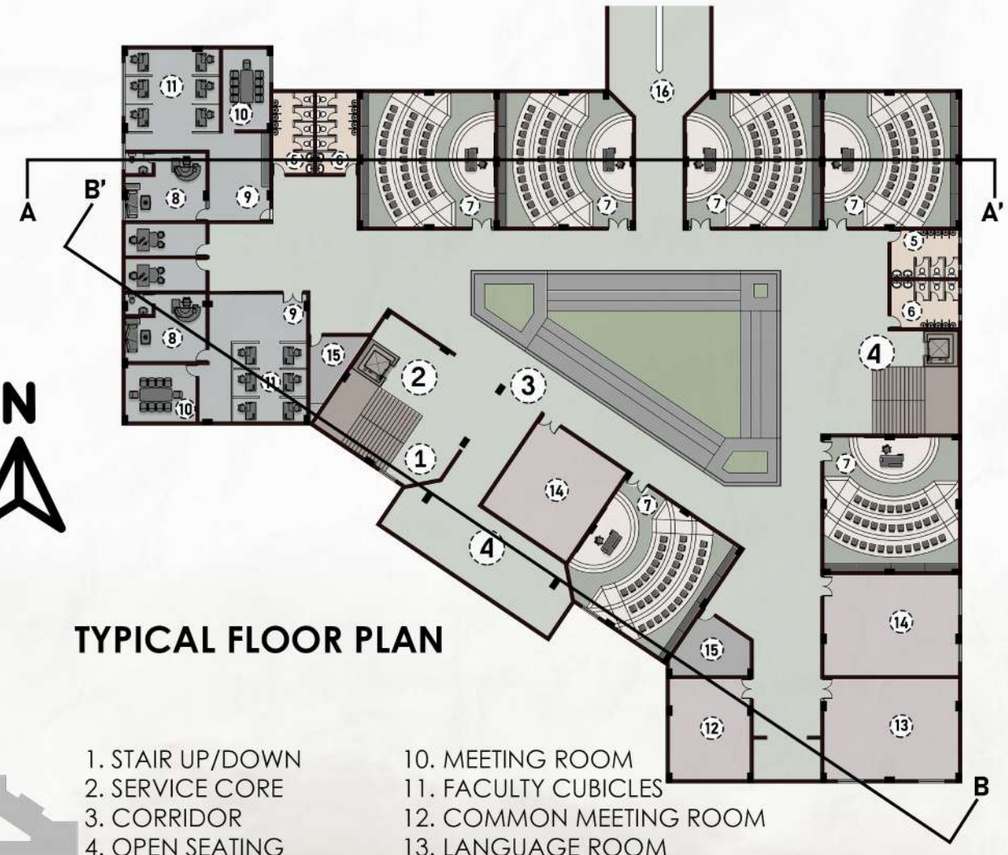


POST - GRADUATE Floor Plans



GROUND FLOOR PLAN

1. MAIN ENTRY/EXIT
2. ENTRANCE LOBBY
3. CENTRAL COURTYARD
4. SERVICE CORE
5. MALE TOILETS
6. FEMALE TOILETS
7. LECTURE HALL
8. DEAN'S OFFICE
9. FACULTY AREA ENTRANCE
10. MEETING ROOM
11. FACULTY CUBICLES
12. COMMON MEETING ROOM
13. LANGUAGE ROOM
14. TUTORIAL ROOM
15. STORE ROOM
16. SECONDARY ENTRY/EXIT



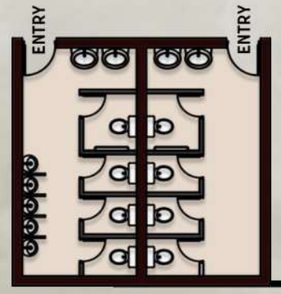
TYPICAL FLOOR PLAN



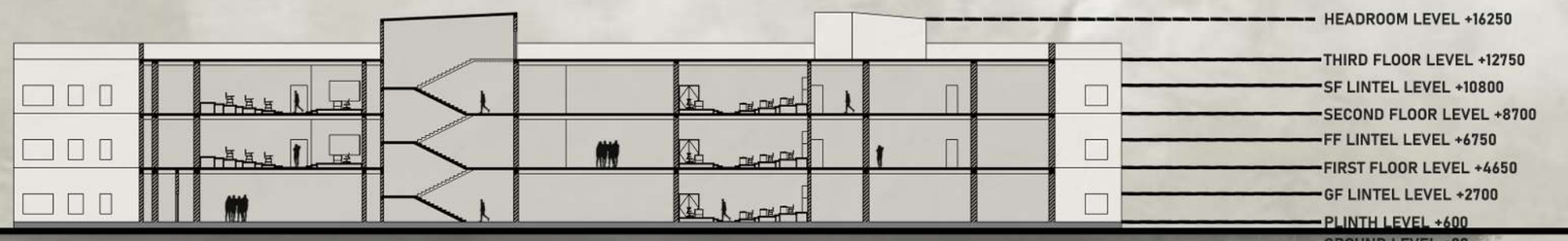


KEY PLAN

1. STAIR UP/DOWN
2. SERVICE CORE
3. CORRIDOR
4. OPEN SEATING
5. MALE TOILETS
6. FEMALE TOILETS
7. LECTURE HALL (TYPE 1)
8. DEAN'S OFFICE
9. FACULTY AREA ENTRANCE
10. MEETING ROOM
11. FACULTY CUBICLES
12. COMMON MEETING ROOM
13. LANGUAGE ROOM
14. TUTORIAL ROOM
15. STORE ROOM
16. CONNECTING BRIDGE



REFINED TOILET DETAIL (BARRIER FREE DESIGN)



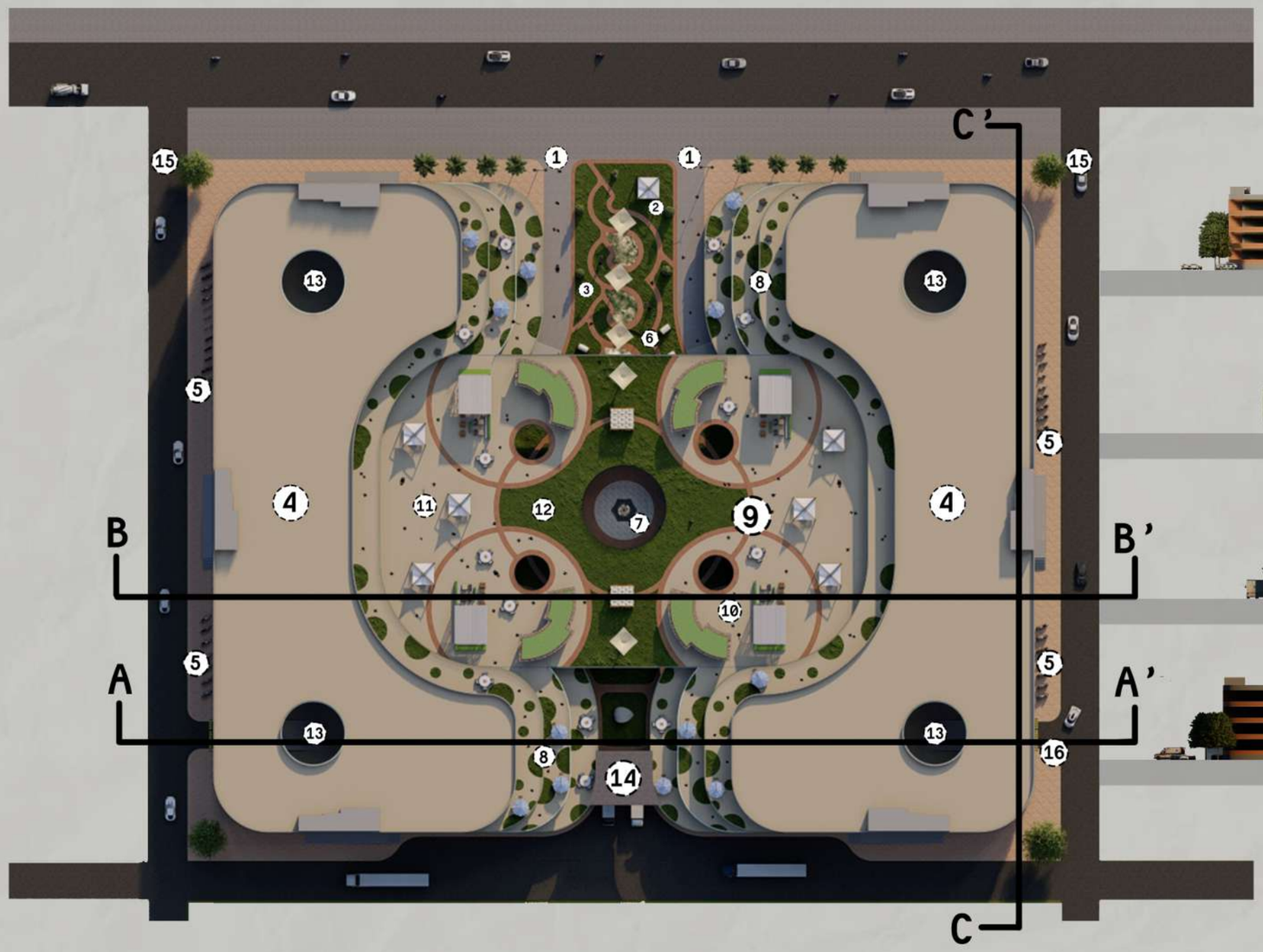
SECTION BB'

Market Redevelopment

LOCATION - BHUBANESWAR
SITE AREA - 20700 SQ.M/ 5.11 ACRES
SEMESTER - 8th SEM
ABOUT - The Unit I Market in Bhubaneswar stands as a vital urban landmark, deeply embedded in the city's post-independence planning history. Conceived as part of Otto Königsberger's neighbourhood unit model, the market was designed to serve local communities with daily necessities while fostering a sense of social cohesion. Over time, it has evolved into a bustling commercial hub that attracts people from across the city, known for its affordable prices and diverse range of goods. Despite its vibrant character, the market now faces challenges related to outdated infrastructure, congestion, and the pressures of rapid urbanization - highlighting the urgent need for a sensitive yet forward-looking redevelopment approach.



Site Plan



ELEVATIONS



NORTH-EAST ELEVATION



NORTH-WEST ELEVATION



SOUTH-EAST ELEVATION



SOUTH-WEST ELEVATION

AREA STATEMENT

1. SITE AREA - 20,700 SQ.M/ 5.11 ACRES
2. GROUND COVERAGE - 13,394 SQ.M = 64%
3. TOTAL BUILT UP AREA - 49,114 SQ.M
4. SKY DOCK AREA - 4,830 SQ.M
5. F.A.R - 2.37

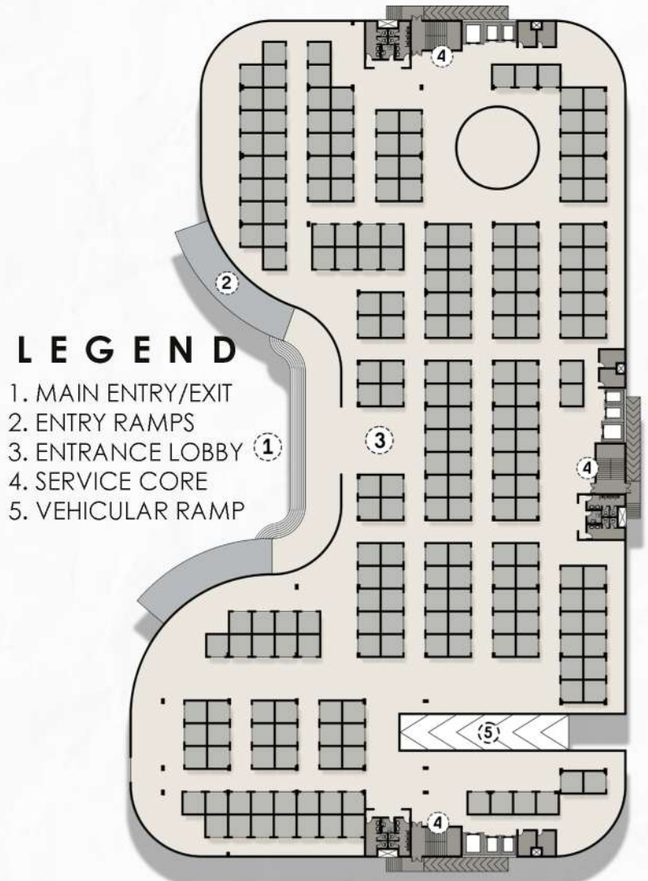
L E G E N D S

1. MAIN PEDESTRAIN ENTRY
2. GAZEBO AND GARDEN
3. LANSCAPE
4. MARKET BUILDING
5. ON SITE PARKING
6. WOODEN PAVEMENT
7. WATER FEATURE
8. TERRACE GARDENS
9. SKY DECK
10. RESTAURANT/CAFES
11. OPEN SEATING WITH SHADES
12. ROOF-TOP GARDEN
13. BUILDING ATRIUM
14. LOADING - UNLOADING DOCK
15. SECONDARY VEHICULAR ROAD
16. VEHICULAR RAMP ENTRY (BASEMENT)

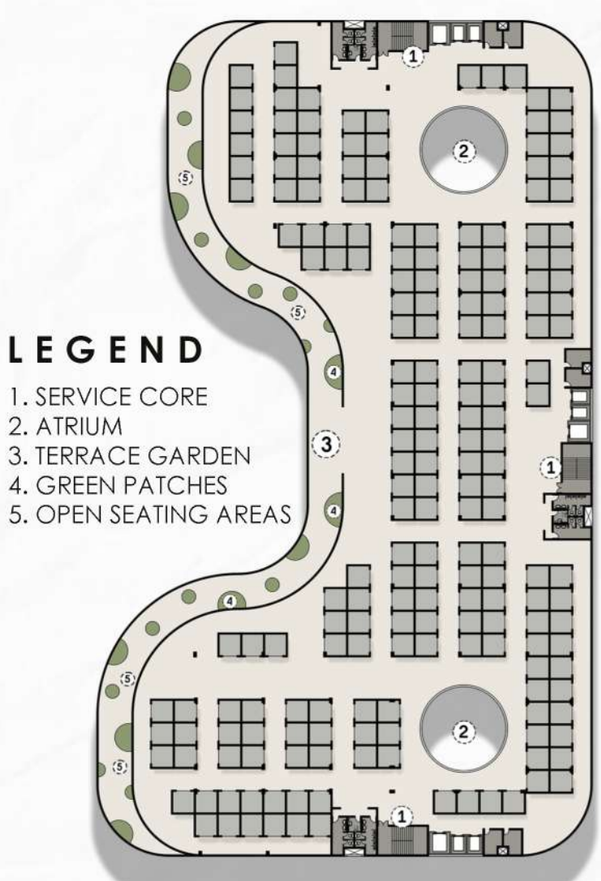




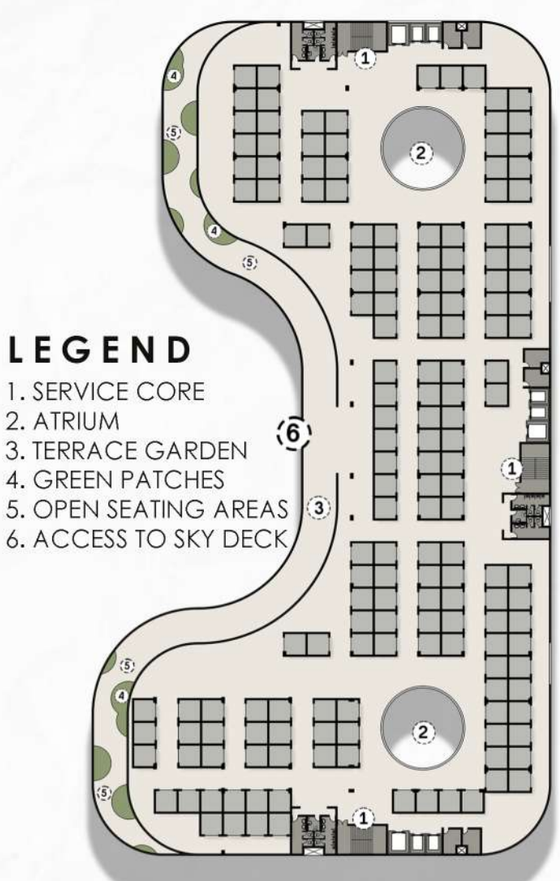
Floor Plates



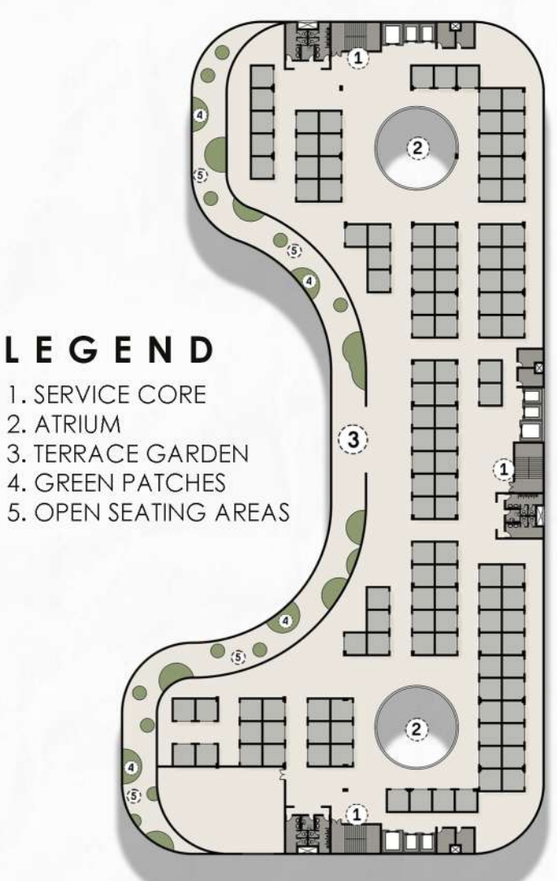
TYPICAL GROUND FLOOR



TYPICAL FIRST FLOOR



TYPICAL SECOND FLOOR



TYPICAL THIRD FLOOR

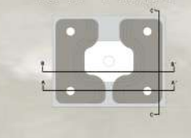


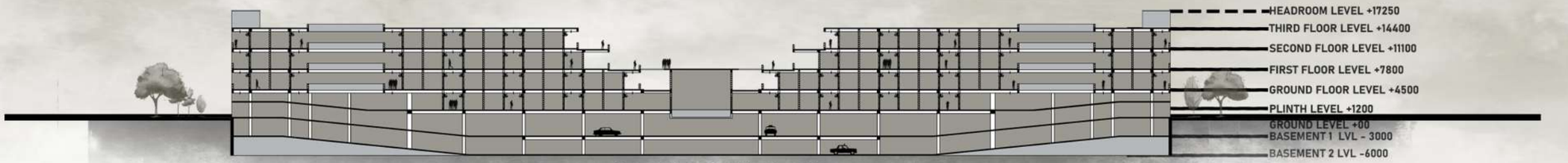




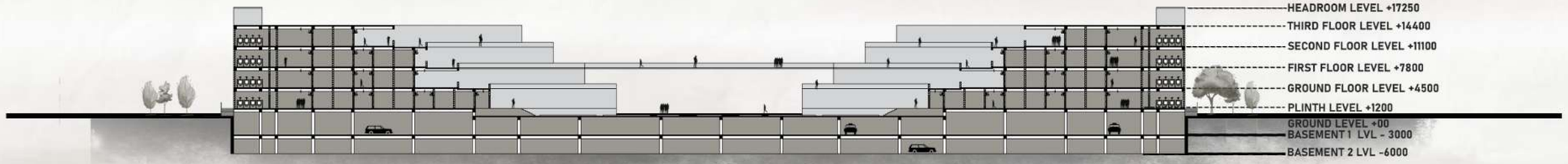
Sections

KEY PLAN

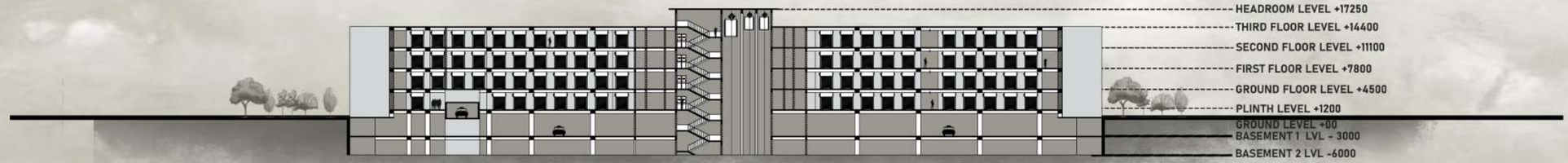




SECTION AA'



SECTION BB'



SECTION CC'







Evolve.io

LOCATION - BHUBANESWAR
SITE AREA - 4000 SQ. FT.
SEMESTER - 7th SEM
ABOUT - The interiors of an architectural firm reflect not only its functional requirements but also its design philosophy and creative identity. As a space that fosters innovation, collaboration, and precision, the interior layout is meticulously planned to balance aesthetic appeal with operational efficiency. From open-plan workstations and material libraries to client meeting zones and model-making areas, every element is designed to inspire creativity while supporting the diverse workflows of architects and designers. The careful selection of materials, lighting, furniture, and spatial organization conveys the firm's ethos and often serves as a live portfolio, showcasing its attention to detail, commitment to sustainability, and mastery of spatial design.



Moodboard CONCEPT AND FURNITURE



CONCEPT

The interior concept emphasizes minimalism with clean lines, neutral colors, and open planning to foster creativity and functionality. A neutral palette of whites, greys, and natural textures ensures a calm ambiance. Large windows, skylights, and a double-height cutout maximize natural light and ventilation, creating a bright, airy, and inspiring workspace that balances openness with defined zones.

FURNITURE SELECTION

- Modular desks with clean lines and natural wood or metal finishes.
- Ergonomic chairs in neutral tones for comfort and simplicity.
- Built-in storage integrated seamlessly into walls and workstations.
- Streamlined meeting tables and benches with durable materials.
- Minimal soft furnishings with subtle textures for warmth.
- Functional, uncluttered furniture enhancing the open and airy layout.

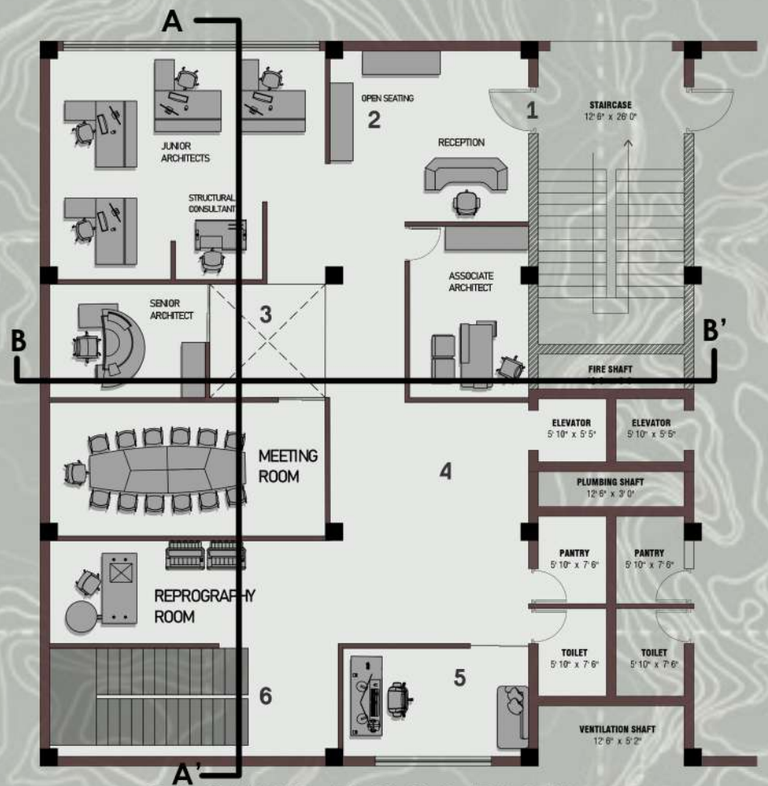
COLOR PALATTE



The minimal color palette emphasizes neutral tones like whites, greys, and beiges, complemented by natural textures like wood and stone. Subtle accents in muted shades add warmth, creating a calm, balanced, and timeless aesthetic.

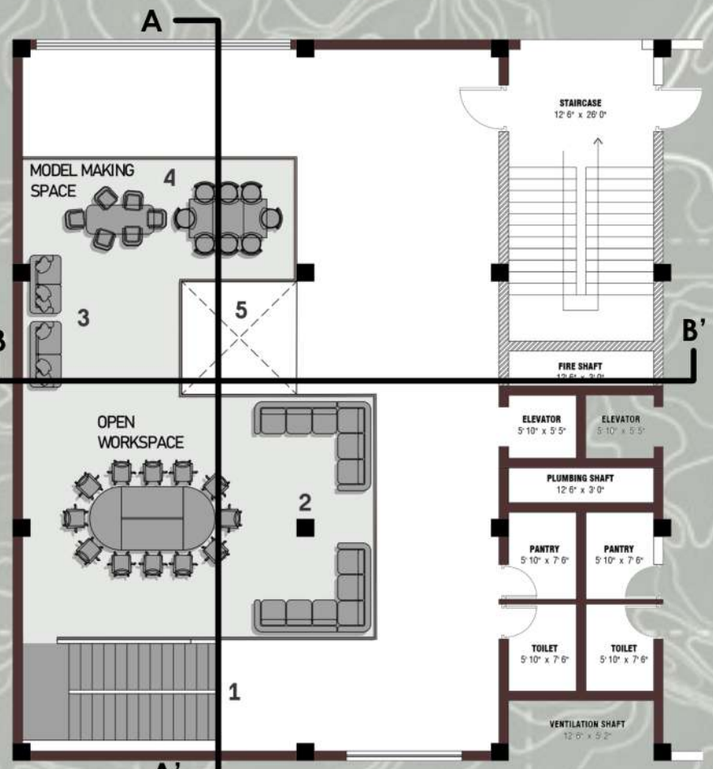


Floor Plans



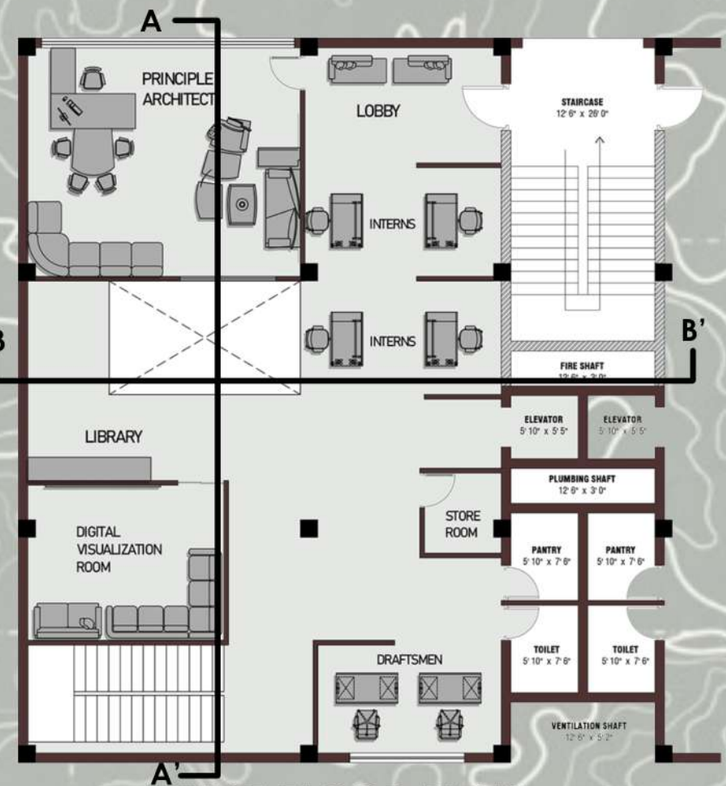
GROUND FLOOR PLAN

1. ENTRY
2. LOBBY AREA
3. DOUBLE HEIGHT CUTOUT
4. SERVICE BUFFER ZONE
5. GAMING ROOM
6. STAIRCASE



MEZZANINE FLOOR PLAN

1. ENTRY STAIRCASE
2. SEATING AREA
3. BOOK STORAGE AND SEATING
4. WNIDOW FACING MODEL MAKING SPACE
5. DOUBLE HEIGHT CUTOUT



FIRST FLOOR PLAN

1. ENTRY FROM 1ST FLOOR
2. OPEN SEATING
3. DOUBLE HEIGHT CUTOUT
4. LIBRARY SEATING SPACE
5. DRAFTSMEN ROOM
6. STAIRCASE

SECTION AA'



SECTION BB'









RENDERED FLOOR PLANS

Views



LIVING SPACE (F.F.)



MAIN ENTRY (G.F.)



INTERNS (F.F.)



MEETING ROOM (G.F.)



MODEL MAKING ROOM (MZN)



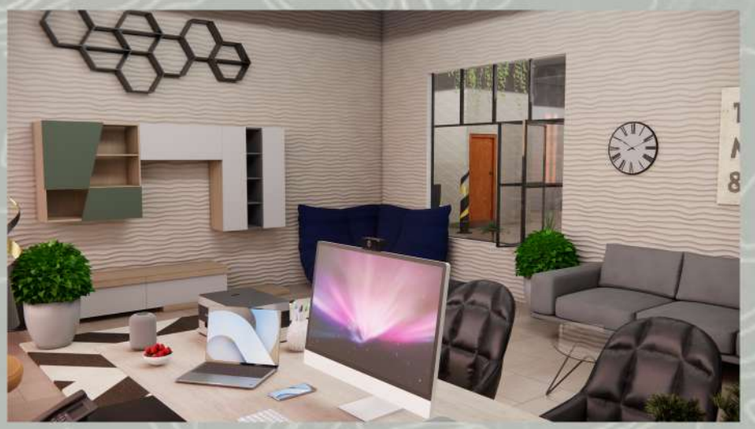
LIBRARY (F.F.)



GAMING ROOM (G.F.)



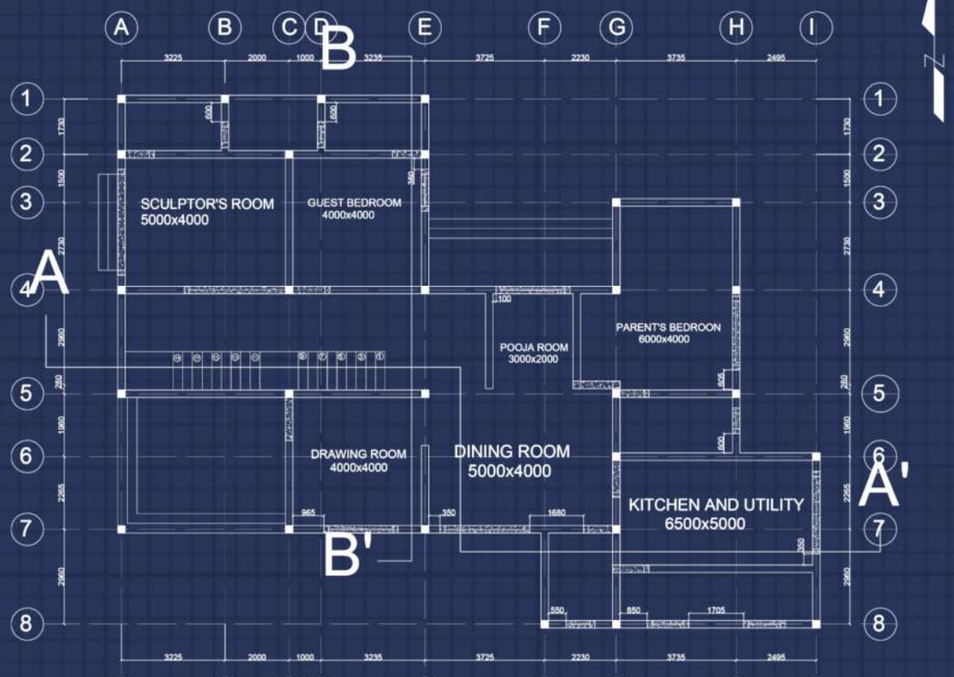
PRINCIPLE ARCHITECT VIEW 1



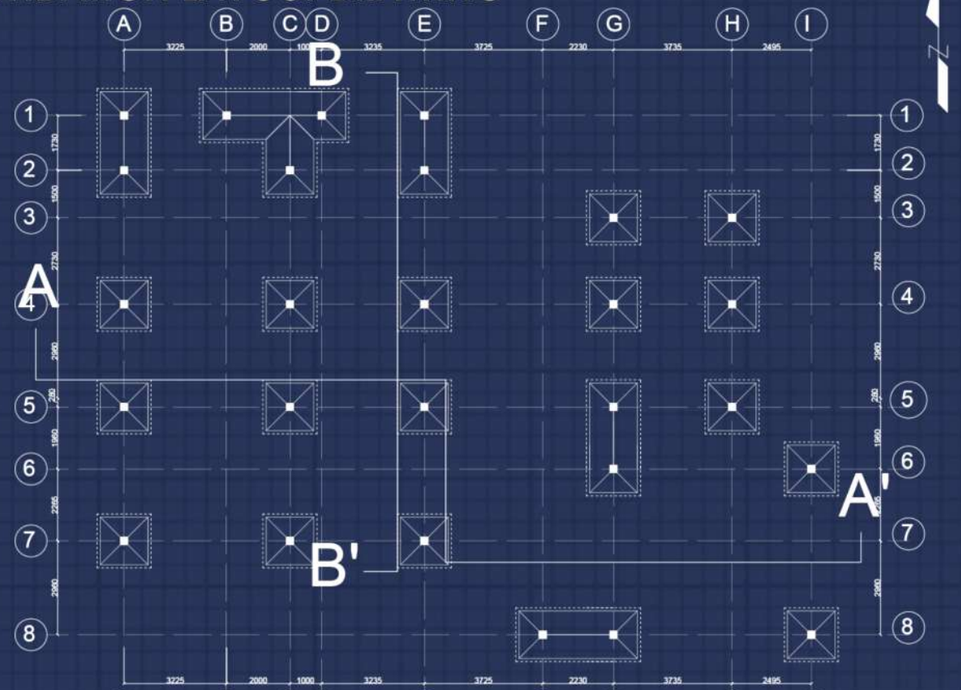
PRINCIPLE ARCHITECT VIEW 2

Nirmaan ATULYAM 3rd SEM - 5TH SEM WD

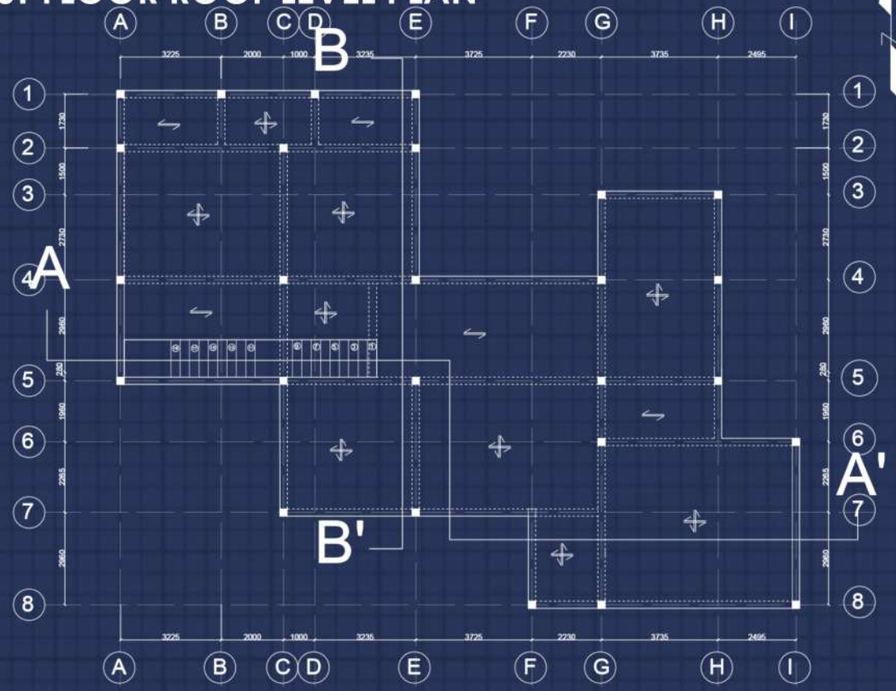
LINTEL LEVEL PLAN



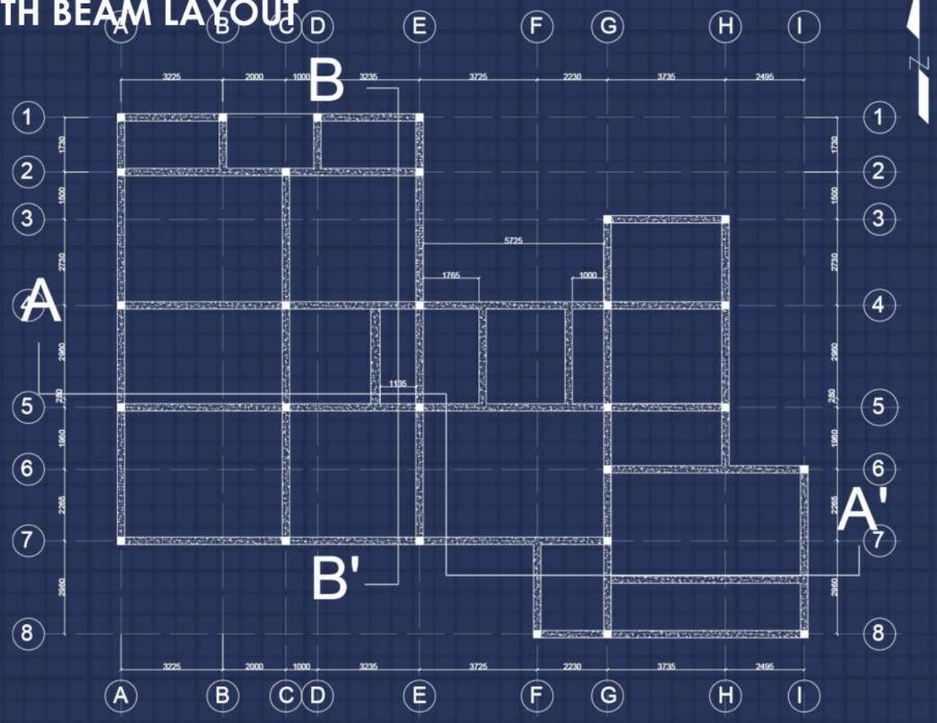
FOUNDATION LAYOUT DRAWING



FIRST FLOOR ROOF LEVEL PLAN

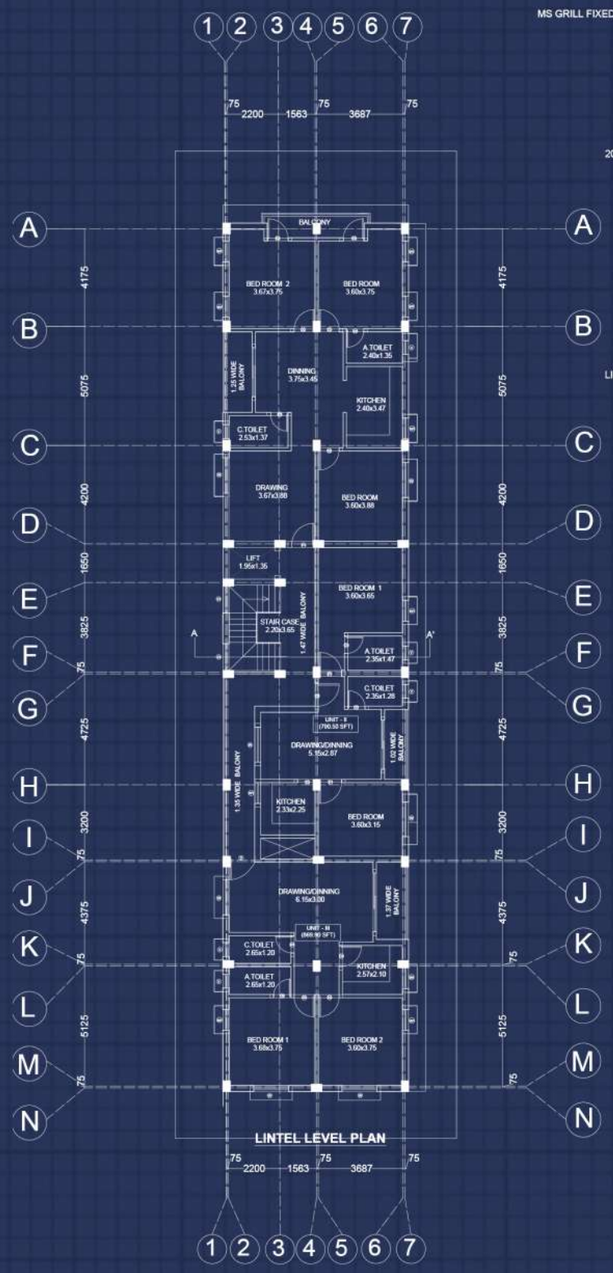


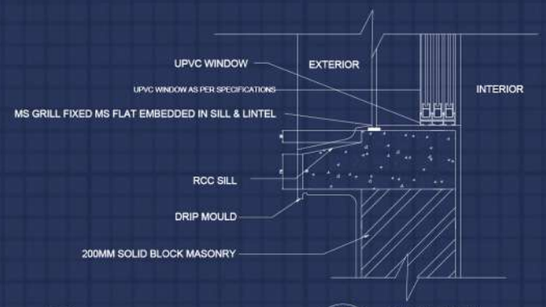
PLINTH BEAM LAYOUT



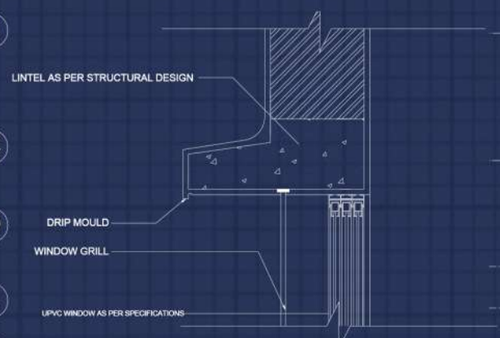
Nirmaan KALINGA APARTMENTS - 6TH SEM WD

LINTEL LEVEL PLAN

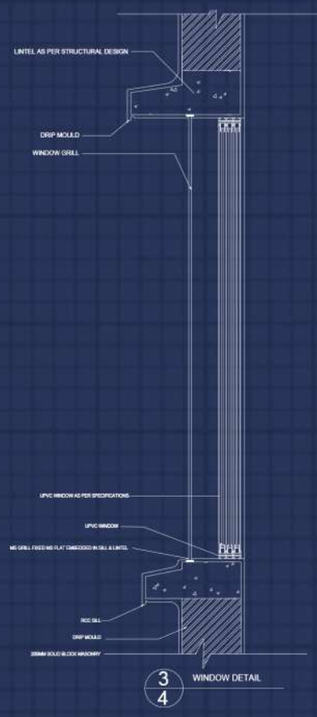




1/4 DETAIL AT SILL

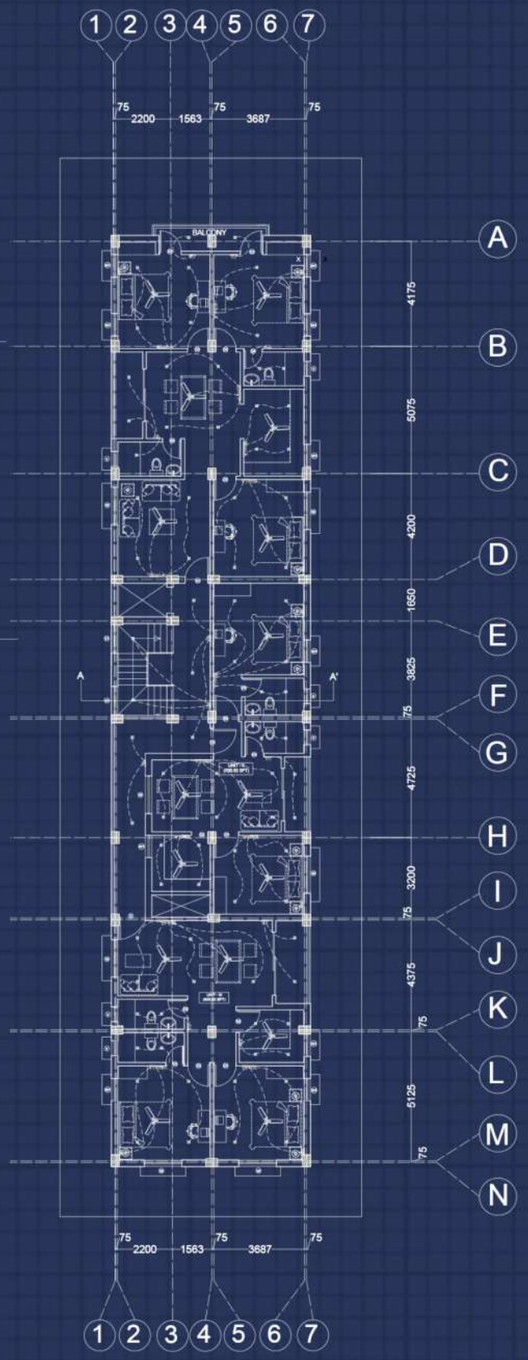


2/4 DETAIL AT LINTEL

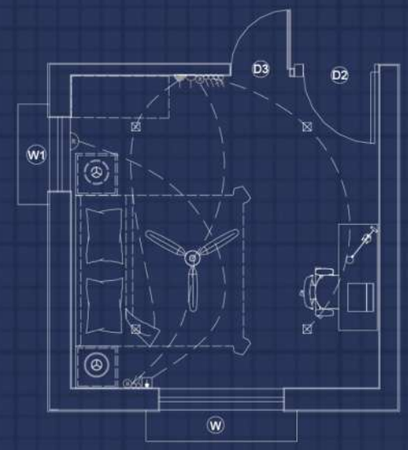


3/4 WINDOW DETAIL

ELECTRICAL LAYOUT DETAIL



11/3 ELECTRICAL LAYOUT



10/3 BEDROOM ELECTRICAL LAYOUT

DETAILS OF LIGHT FIXTURES AND MISC. ITEMS

#	Legends	Description	Watts	Mounting height from FFL to the bottom	Remarks	elevation
1.0	Receptacles					
1.1		6A Switch socket outlet (single)	100W	As per Elevation	-	
1.2		6A Switch socket outlet (twin)	200W	As per Elevation	-	
1.3		6A/16A Switch socket outlet	500W	As per Elevation	-	
1.4	(FP)	16A Socket Outlet for Geyser	1500W	As per Elevation	25 LTRS	
1.5	FP W/P	16A Weather Proof Socket Outlet for Geyser	1500W	As per Elevation	25 LTRS	
1.6		Ceiling rose	-	As per Elevation	-	
1.7		Telephone Outlet	-	As per Elevation	RJ 11 Type	
1.8		Intercom Outlet	-	As per Elevation	RJ 11 Type	
1.9	TV	TV Outlet	-	As per Elevation	Coaxial Type	
1.10	Net	Cat6 connection for IPTV	-	As per Elevation	-	
1.11	HDMI	HDMI		As per Elevation		
1.12	WM	16A Switch Socket outlet	500W	As per Elevation	W/M : Washing M?C SSO Near Utility	
1.13		40A,4P Isolator in sheet steel Enclosure (raw power for AC)	-	As per Elevation	For A/C System -OutdoorUnit	
1.14		16A one way switch with indicator (geyser)	-	As per Elevation	For A/C System -OutdoorUnit	
1.15	VM	Video Monitor with Handset	-	As per Elevation		
1.16	VDP	Camera for Video Door Entry	-	As per Elevation		
1.17		Fan Regulator	-	As per Elevation		
1.18		Fan Regulator	-	As per Elevation		
1.19		6A One way/ Two way switch	-	As per Elevation		
1.20		40A,4P Isolator in sheet steel Enclosure (raw power for AC)	-	As per Elevation	For A/C System -OutdoorUnit	
1.21	C1	6A/16A Switch Socket outlet	500W	As per Elevation		
1.22	C2	6A/16A Switch Socket outlet	500W	As per Elevation		
1.23	C3 C1 C2 C3	6A/16A Switch Socket outlet	500W	As per Elevation		
1.24	DB	Distribution Board three phase- 4 rows (450mm-w x 750mm-h x 95mm-d		As per Elevation		
2.0		Junction box		As per Elevation		
2.1		Notions (Wherever Applicable)				
2.2	gpo: General purpose outlet , w/p : weather proofing					
2.3	ref: Refrigerator, mw: Microwave , ch: Chimney, R.O.System, gey : Geyser, wm: Washing Machine, dw: Dishwasher, fc: Food cutter, to: Toaster, g/b: Gnder/Blender, Oven: Oven , ex: Exhaust fan					

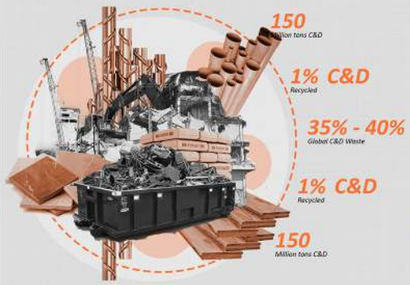
Competetion Works

ANDC 2023

The ANDC Trophy 2023-24 addresses a pressing urban need by exploring the potential of adaptive reuse and sustainable design through the project **"Aalo-Chhaya: A Play of Light and Shade."** The design responds to a lack of pause points and rain shelters along pedestrian routes within institutional campuses, reimagining neglected urban voids into meaningful spaces of rest, interaction, and architectural identity. Rooted in contextual responsiveness and environmental stewardship, the pavilion is constructed using construction and demolition (C&D) waste—from polypropylene cement bags and TMT steel bars to reused PVC pipes—demonstrating how architectural design can innovatively repurpose urban waste into functional, aesthetic, and socially inclusive structures. Drawing from modular geometry, particularly the hexagon for its structural and spatial efficiency, the project embodies principles of flexibility, scalability, and community-focused usability. The Aalo-Chhaya pavilion stands as a testament to architecture's power to transform overlooked materials and spaces into resilient micro-urban interventions, promoting environmental consciousness and social cohesion in the built environment.

01 DESIGN SHEET

AALO - CHHAYA
PLAY OF LIGHT AND SHADE



Construction and Demolition waste (C&D)

Construction and demolition (C&D) waste is generated from construction, renovation, repair, and demolition of houses, large building structures, roads, bridges, piers, and dams.
C&D waste is made up of wood, steel, concrete, gypsum, masonry, plaster, metal, and asphalt.

India's Major concern

Construction and demolition waste is a major environmental concern in most of the Indian municipalities, as the

- **CW generated** in India is **150 MT** and
- **Accounts** for **35%–40%** of the **global C&D waste** annually
- **Recycles** only **1%** of the C&D waste that is produced,
  - According to Center for Science and Environment estimates

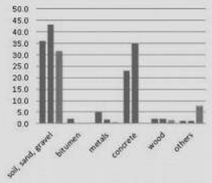


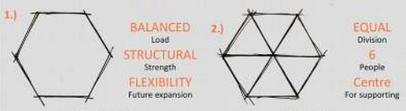
Table shows the percentage of waste in construction according to different surveys in india.

WASTE COMPOSITION

In C & D waste debris was more along with that metal and concrete are also takes major portion in waste.
Wood and other waste takes small part due to high reuse value.

Components of C & D waste	TIFAC (%)	MCD Survey (%)	Survey IL & FS (%)
soil/sand, Gravel	36.0	43.0	31.5
Bitumen	2.0	-	-
Metals	5.0	-	0.4
Concrete	23.0	35.0	-
Wood	2.0	2.0	1.5
Others	1.0	1.0	7.6

FORM DERIVATION:



HEXAGON
Due to balanced distribution and structural strength hexagonal shape was opted.

DIVISION
For accomodation of Six people hexagon was equally divided into 6 wedges.

PROVISION
For shading, cantilevered extension was Provided alternate.

RAISING
By raising the plinth final structure was erected

SITE SELECTION



Site was proposed according to parameters such as

- No of users using the path
- Type of users
- Distance between the blocks
- Space without obstruction

MAJOR USERS

- Students
- Faculty
- Staffs
- Visitors
- Drivers

- The location is along the side of the main road of campus
- Absence of resting places in terms of shaded and non-shaded areas along the entire spine of the 1.7km road
- An interactive space and especially as recreational & resting.
- Total footfall of approximately 2360 which includes the elderly person in the campus.
- Space would be very significant as well as beneficial for Users

MATERIAL USED



POLYPROPYLENE BAGS

India - 2ND largest **producer** of **cement**. Along with the cement ,**polypropylene bags** are also produced.
- **Cement bags** becomes a **waste** after construction of building
- India **generates** around **300,000 tons** of polypropylene bags,**annually**



STEEL REBARS

India - 2ND largest steel producing country in the world. Bended rebars and small pieces after cutting are thrown as waste.
- **Steels** accounts for **7-10%** in construction waste
- India **generates** around **300,000 tons** of polypropylene bags



PVC PIPE

India currently **produces 1.555 mn t/yr of PVC.** The country also **imported** an average of **1.86mn t/yr** of PVC from 2018 to 2021, according to GTT data.
- **Pipes** with small holes or with bends are thrown as waste in construction



DEBRIS

India generates over **62 million tons (MT)** of waste in a year
- **30 %** of it is **recycled**
- Reusing the discarded structural members for creating something new structure

REUSING

- Other materials such as GI wires, cement and aggregates are used as per needed in a negligle amount
- Huge debris blocks are used for constructing sitting spaces and plastered to get rid of uneven surface
- Vegetation by the means of creepers and hanging pots are used to make the space more lively

COSTING TABLE

SR No.	TYPE OF MATERIAL	TOTAL QUANTITY	MEASURING QUANTITY PER UNIT	TOTAL QUANTITY IN MEASURING UNIT	COST PER MEASURING UNIT (in Rs)	Approx. ESTIMATED COST (in Rs)	ACTUAL COST (in Rs)	REMARK
1	32 mm dia rod	3 m	8.03 kg per metre	24.09 kg	71	1,710.39 = 1,750	0	Rod was collected from the University construction scrapyard
2	12 mm dia rod	16.45 m	2.12 kg per metre	34.87 kg	60	2,092.2 = 2,100	0	Rod was collected from the University construction scrapyard
3	8 mm dia rod	5.4 m	0.60 kg per metre	11.45 kg	58.5	681.27 = 700	0	Rod was collected from the University construction scrapyard
4	Electrical pipes	13.7 m		44.9 ft	12	538.8 = 540	0	Broken pipes were obtained from construction site inside University campus
5	Cement bags	60	0.76 sq. m	45.6 sq. m	7.4	344.56 = 350	0	Discarded bags obtained from construction site inside University campus
6	Slag cement	85 kg		85 kg	7.5	637.5 = 640	0	Collected from construction site inside University campus
7	Aggregate	4x quantity of slag cement		340 kg	1	340	0	Collected from construction site inside University campus
8	Sand	2x quantity of slag cement	0.103 cu m	170 kg	892	92.75 = 93	0	Collected from different areas inside University campus
9	Tiles	1.3 sq. m		1.3 sq. m	70	103	0	Waste damaged tiles from scrapyard
10	Plastic rope	20 m		20 m	0.42	8.44 = 10	0	Plastic Rope were collected from the University construction scrapyard
11	Concrete block	12 x 10.2m x0.2m x0.4m		0.2088 cu m	2,033	815.9 = 820	0	Damaged and rejected Pre cast concrete blocks were collected from construction site inside university campus
12	Needle and thread					70	0	Needle and thread were collected from the Material Lab of the Architectural Department
13	GI wires	5 m		5 m	4	20	0	GI wires were collected from the University construction scrapyard
14	Plastic Bottles	40		40	25	1000	0	Collected by the means of a Plastic collection drive from the students
15	Jute thread	4.5 kg		4.5 kg	150	715	0	Collected from scrapyard
TOTAL COST						8,775 = 8,800	0	

USABILITY

- Acts as an sun shade in summer and provides a pause point to sit and rest.
- Interaction space for students for leisure, Relax zone with comfort.
- Venues for cultural and art events, like art exhibitions and performance.
- Acts as an Rain shelter,waiting during raining and monsoon season.
- Used for education purposes, such as outdoor classes, workshops, and lectures.
- Open and airy design allows for a connection with nature.
- Architectural attraction,provide uniqueness to the area.
- Space offered by pavilion suits for meditation and yoga practices.

STAGES OF CONSTRUCTION

FOUNDATION

Substructure
Foundation was laid by the usage of slag cement along with reused Rebars welded together.



Excavation
Excavating the pit and a layer of sand is compacted on it.



Framework
TMT Bars placed in its respective positions and PCC Bed was laid.



FRAMEWORK

PVC pipes
TMT bars are inscribed inside PVC pipes to prevent further corrosion of bars.



Roof
Construction of structural frame by welding together at joints.



Sewing
Sewing of waste polypropylene cement bags (cleaned & washed)



CONSTRUCTION

External skin
Used cement bags are sewed and attached to structural Frame with G.I wires.



Benches
Broken ACC blocks are arranged and joined together to form seating benches.



Mosaic Flooring
Broken granite pieces laid on PCC bed to form a mosaic pattern.





ANDC TROPHY 2023-24

Registration code 66ANDC -150

NASA

02 DESIGN SHEET

AALO - CHHAYA
PLAY OF LIGHT AND SHADE



PROBLEM

There is a **lack** of **Pause point** and **Rain shelter** on a long stretch of 820m from University main gate to academic building .
- Users especially **Students** are feeling **difficulty** on the **way** due to absence of resting space.

VISION

To construct a **prototype pavillion** inside the university which gives the **path users** a **resting space** and also act as a **point of interest** for the **visitors.**
To **provide** of a interesting space for interaction and recreation.

"Make it simple, but significant"



TIMELINE

EXCHANGE OF KNOWLEDGE
- Breaking down the brief
- Analysing the C&D waste
- Brainstorming ideas
- Research and inspiration

THINKING MADE VISUAL
- Identifying Problems and Finding Solutions
- Conceptualising pre-design
- Converting ideas into **Reality**

C&D WASTE TO WEALTH
- Analysing available C&D waste
- Collection of materials
- Treatment of materials

IMAGINATION TO EXECUTION
- Preparation of site
- Raisal of structure
- Welding and connection
- Roofing and finishing

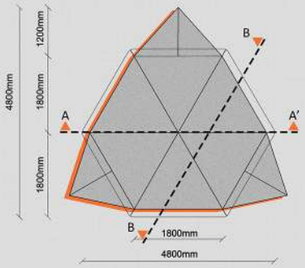
OBSERVING *"AALO CHHAYA"*
- Preferred to visit
- Used to sit and talk
- Stone pathway was made
- Using as photo-booth

STAKE HOLDERS



SITE

a The site is located in respective university, on the curb of main pathway connecting all major building blocks, acting as a focal point.
a The site provides scenic views towards east & west sides.



PLAN

DESIGN PARAMETERS

The **Prototype Pavillion** is conceptualized to accommodate 6 people & the Usability of the space is mostly resting and interaction by the students.

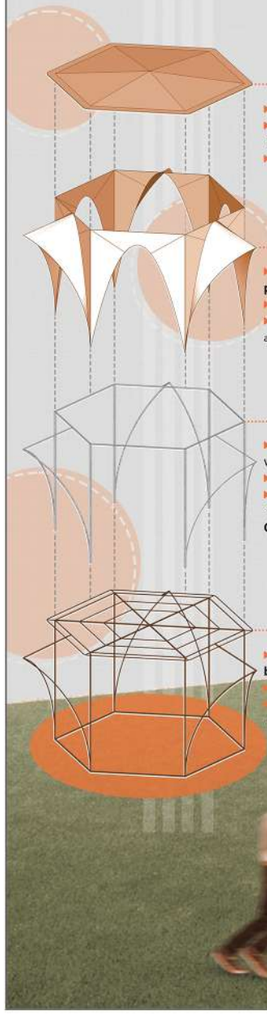
"Balanced distribution,Structural strength Flexibility and scalability"

- Design contains **6 wedges** for **6 people,** derived the form of a hexagon
- **Hexagon** shape **utilizes** its **area** efficiently

SUSTAINABILITY

- Usage of waste materials leads to sustainability
- The structure is energy efficient constructed with minimal usage of new materials
- Structure helps to improve social cohesion

AXONOMETRIC BLOWUP



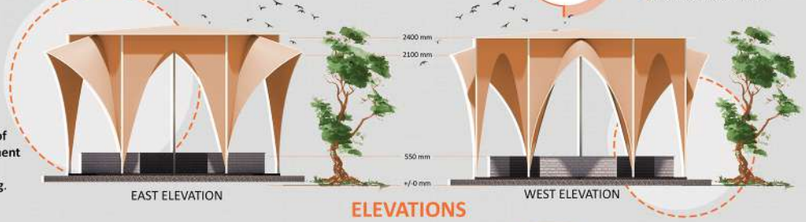
ROOF
- **100mm** Rised **Sloppy Roof**
- **Used Polypropylene Cement Bags**(sewed).
- Raisal **prevents Water-log.**

SKIN OF PAVILLION
- Made by **sewing** used **polypropylene cement bags**
- Act as **walls**
- **Waterproof**

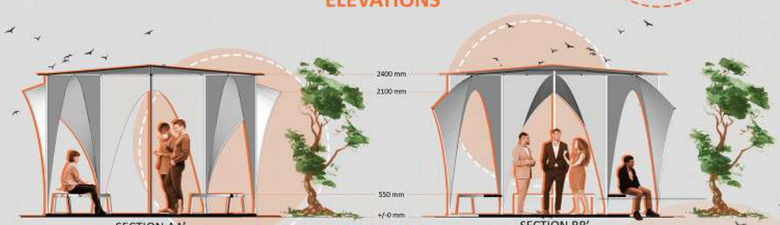
a **Sustain** for a **long time**

PVC PIPES
- Reinforcement is **covered** with used **PVC pipes**
- **Aesthetically** pleasing
- PVC Pipes also **protect** main frame from **Weathering Conditions** & **Corrosion**

REBAR WIREFRAME
- Small pieces of waste **TMT bars**
- **Framework**
- **Welded** together to form main **Frame structure**



ELEVATIONS



SECTIONS



ANDC TROPHY 2023-24

Registration code 66ANDC -150

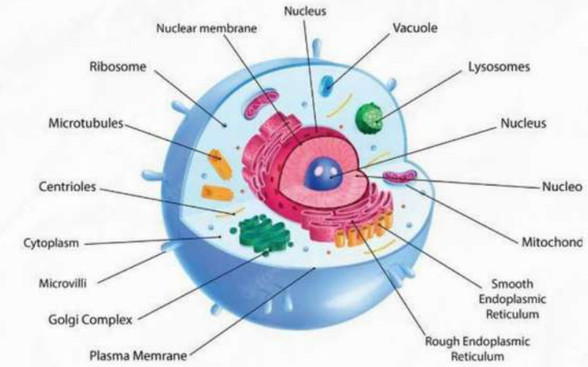
NASA

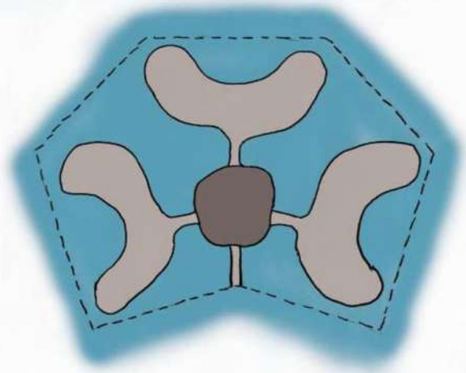
Competetion Works

VERTICAL DESIGN STUDIO 2023

The Vertical Design Studio 2023 is a collaborative academic exercise that brings together students from diverse levels of architectural education to engage in an intensive design process. This studio provides a platform for experimentation, interdisciplinary thinking, and contextual responsiveness. Group A2, consisting of Soham, Ashutosh, Srushti, Vishnupriya, Kalyan, and Swabhiman, explored design solutions grounded in real-world challenges, guided by a critical approach to site, program, and user needs. The project fosters a deeper understanding of spatial strategies, sustainability, and social engagement within the built environment.

CONCEPT



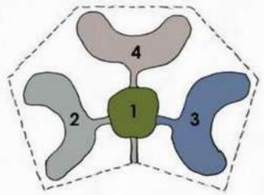


Around 3.8 billion years ago, all the organisms on Earth originated from a **"single cell."** Among the various parts of the cell the **nucleus** is the brain of the cell which functions to provide structure, take in nourishment, convert energy into energy, and carry out specialized functions. All the different parts with specific functions together build a cell.

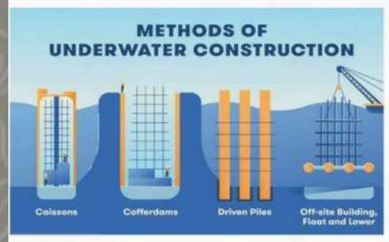
Similar to this, we have conceptualised our futuristic sustainable habitat to symbolise with a human cell, with efficiently zoned out bodies with specialised functions and features

Our habitat is zoned out typically into 4 bodies:

1. SECURITY, MONITORING & SURVEILLANCE CELL, ENERGY CONSERVATION BODY (Nucleus of the habitat)
2. RESEARCH, INNOVATION, INFORMATION DEPARTMENT, FARMING AND GREEN HOUSE
3. RECREATIONAL HUB, HOSPITALITY, LODGING PLATFORM AND TOURISM
4. HOSPITAL, INSTITUTIONAL SPACE AND LIVING PLATFORM



FLOATING MECHANISM







A driven pile is a large, vertical steel column that is hammered into soil or rock by a massive machine.
Similar to a nail being hammered into wood, driven piles create strong foundations after being filled with concrete.

A common of underwater construction involves building off-site, then floating the structure with a barge before lowering it into the water using weights.
The structure can then be attached to a foundation on the seafloor using steel driven piles.

MATERIALS



STYROFOAM

The blocks can be covered externally with brickwork, timber cladding or render and internally with plaster board and plaster.

DRIFTWOOD

From the skeleton of an entire tree to a piece of driftwood the size of a pebble, each piece of driftwood provides a benefit to the beach ecosystem.
Driftwood provides stability to the sandy beach environment.





LUMBER WOOD

Collectively wood that's been processed for use as a building material—wood that has been milled and cut into boards or planks.

Also used in cabinetry, furniture and flooring and pallets.

ADVANCED MATERIALS



ACRYLIC GLASS

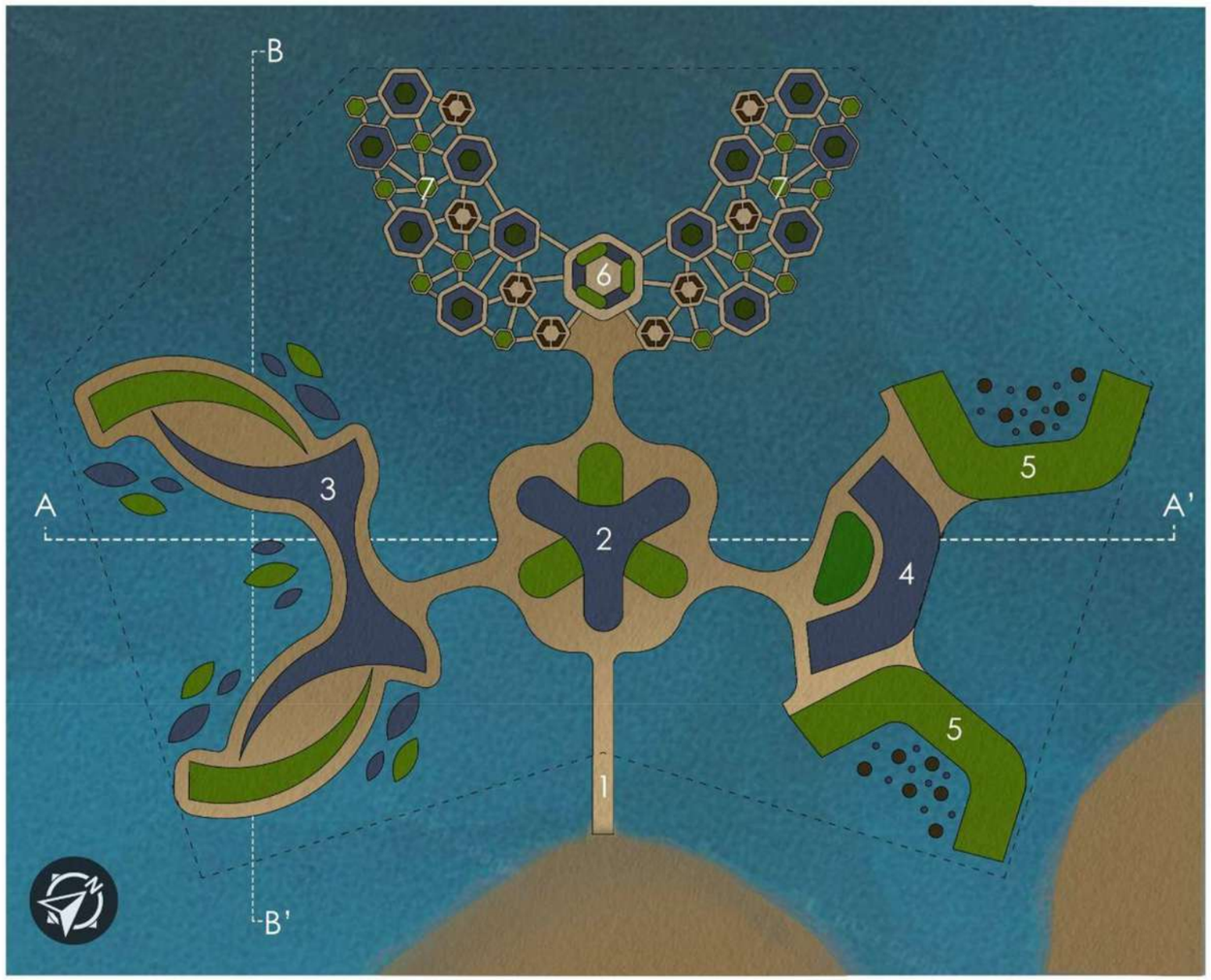
High impact resistance, excellent thermal insulation, and noise reduction properties. Skylights and



BIO ALGAE

This biomass is produced within a "photo bioreactor," or additional transparent outer panels that are attached onto the sides of the building and supply the structure with electricity. These bioreactors are filled with water and pumped with liquid nutrients and carbon dioxide for the algae to grow, as well as to prevent them from rotting. It's these bioreactor panels that generate the energy that powers the structure.

SITE PLAN



LEGENDS

1. MAIN ENTRY

2. SECURITY, MONITORING & SURVEILLANCE CELL, ENERGY CONSERVATION BODY

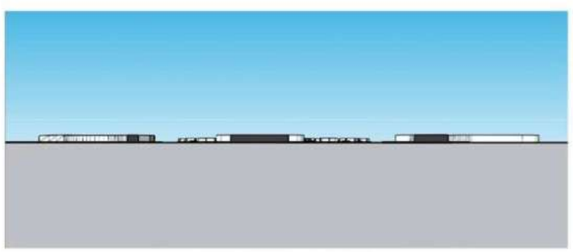
3. RESEARCH, INNOVATION, INFORMATION DEPARTMENT, FARMING AND GREEN HOUSE

4. RECREATIONAL HUB

5. HOSPITALITY, LODGING, TOURISM

6. HOSPITAL, INSTITUTIONAL SPACE

7. LIVING PLATFORM

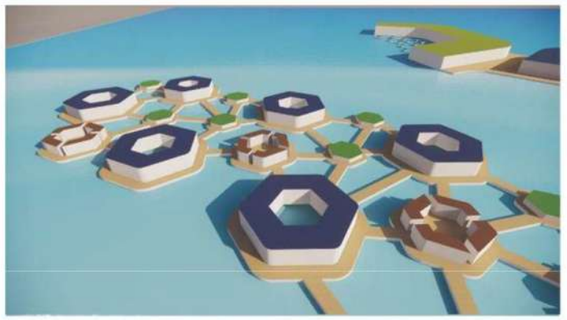


SECTION A-A'

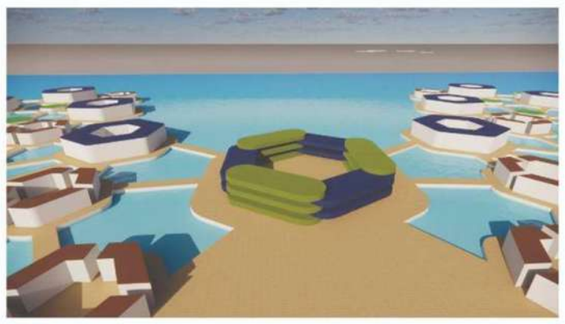


SECTION A-A'

ELEVATIONS & VIEWS







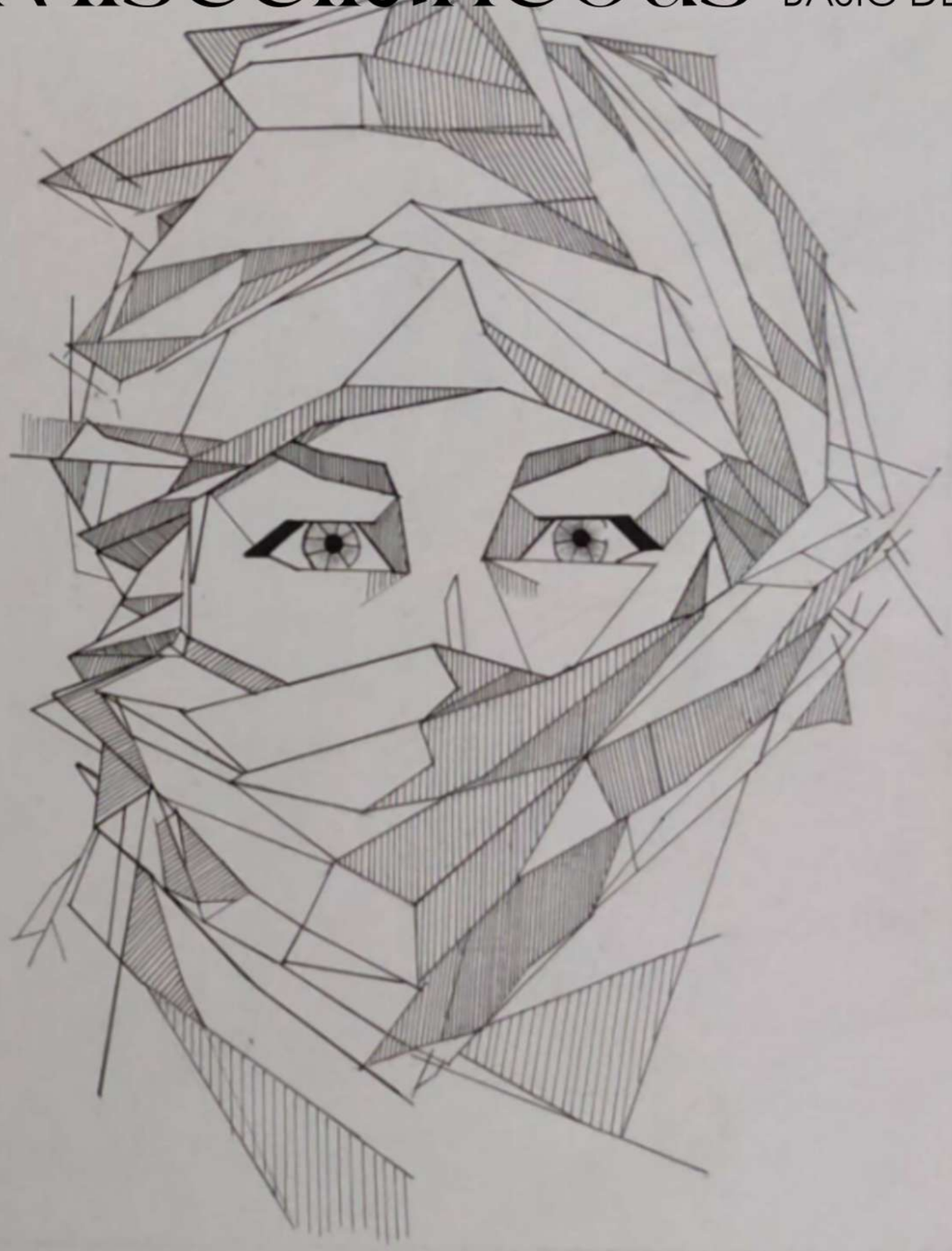








Miscellaneous BASIC DESIGN









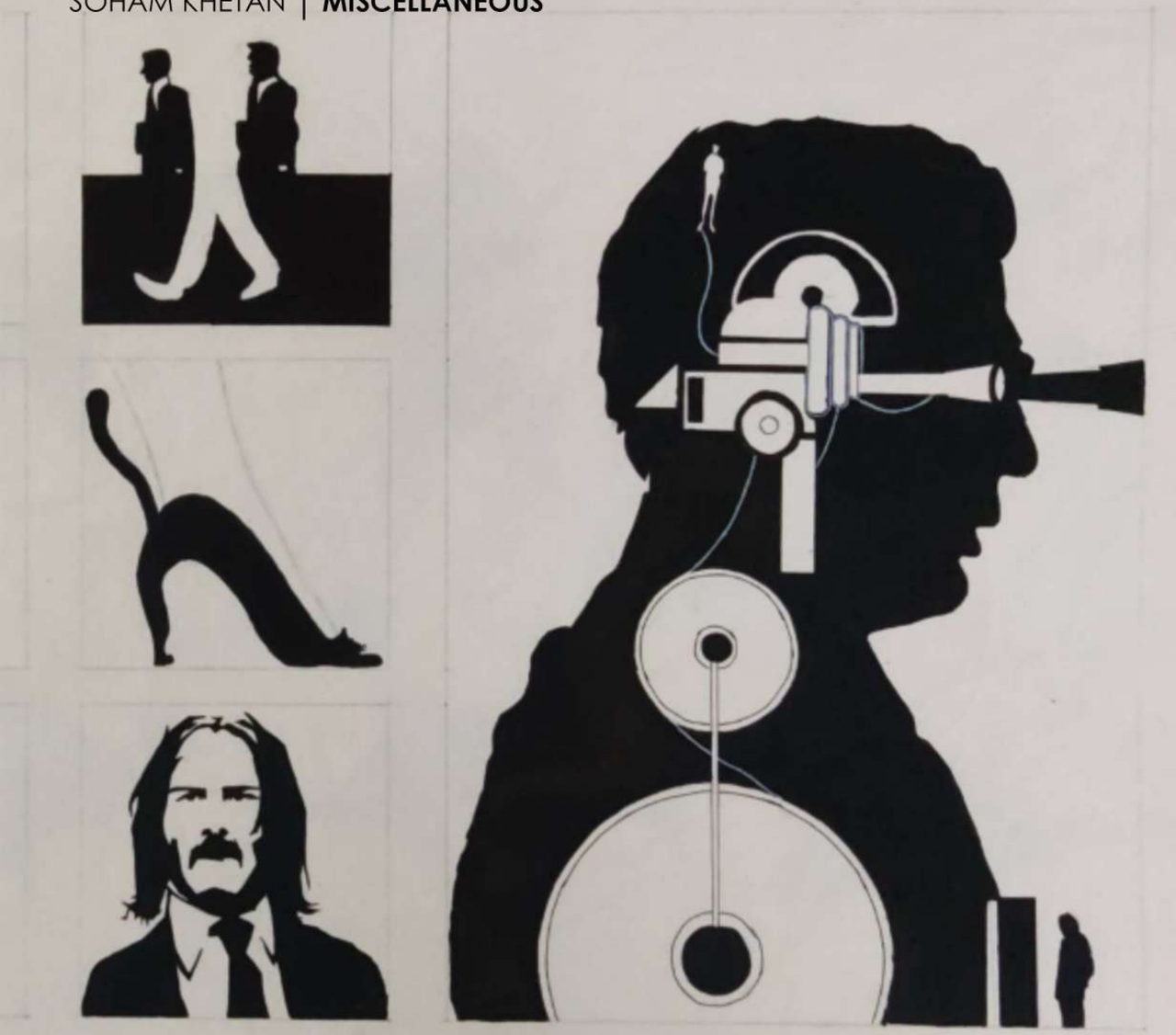


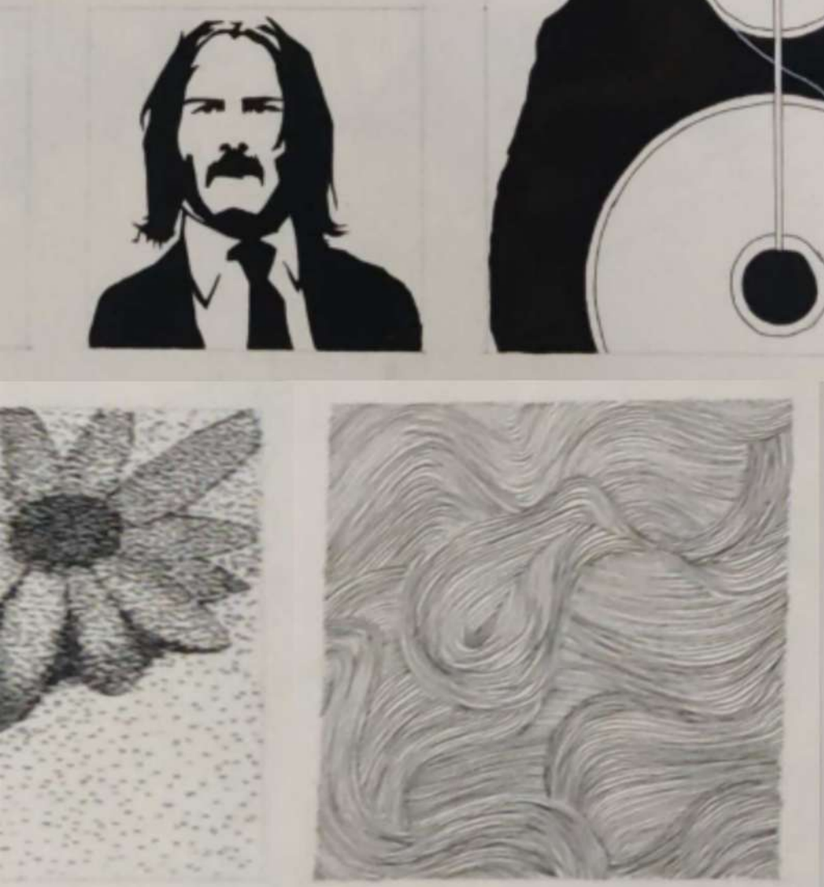
















Thank You

sohamkhetan@gmail.com

+91 9356846922

@soham_ktan

18/10/2003

Chandrapur, Maharashtra

https://issuu.com/sohamkhetan

Soham Khetan